claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf_2a64bc88-2a8\agent-a5a15f1c75a5655fc.jsonl`
- SHA-256: `5dadfd31c0f238da3eb570000f0ef512a59a93cf792f1eb7a6cc7e5fb3857d74`
- Source modified: `2026-06-10T05:40:49+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_2a64bc88-2a8`
- session_id: `45ea3a91-654d-4972-9cd3-65f35db54caf`

Transcript

1. user (2026-06-10T05:33:14.897Z)

```
CONTEXT ? two sibling repos on a Windows machine:
- INDEXER: C:/Users/avidu/Projects/badminton-highlight-indexer ? local AI badminton(->racket-
sports) highlight indexer + rally detector. FastAPI + SQLite + ffmpeg + GPU CV via WSL; Gemini =
paid cloud AI. docs/ tree is rich. Current branch docs/next-steps-multisport (master = main
branch).
- COLLECTOR: C:/Users/avidu/Projects/sports-data-collector ? golden-label
acquisition/curation/ingestion CLI (system-of-record candidate for training corpus).
Project state (June 2026): scaffolding complete; owned-model roadmap agreed
(docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md) but NO platform/owned-model implementation started;
next committed platform slice = async job model. Licensing guardrail: MIT/Apache-2.0/BSD only
for code+data+weights; paid LLMs are inference-only, never training-data sources.
ENVIRONMENT RULES: this dev environment has NO GPU/torch/cv2 ? do NOT attempt to run the video
pipeline or import torch/cv2 modules. Test suites are offline/fully-mocked and SAFE to run. You
are READ-ONLY: never edit/create/delete repo files; running tests and git read commands is
allowed.
QUALITY BAR: cite file paths (and line numbers where possible) for every finding. Report what IS
in the code/docs, not what you assume. Be skeptical and concrete. Your final structured output
is consumed programmatically by an orchestrator ? return raw data, not prose for a human.
TASK: ground-truth the PRODUCT-MARKET-FIT thesis with fresh external research. You have
WebSearch/WebFetch (load via ToolSearch if needed).
First read the internal strategy: docs/PMF_MARKET_RESEARCH.md, docs/COMMERCIALIZATION.md,
docs/ROADMAP.md, docs/NEXT_STEPS.md, docs/PLATFORM_ARCHITECTURE.md (skim for the hosted-API cost
model). Extract the core thesis: who is the customer (Bangalore/India academies? prosumers?),
what is the wedge (upload full match video -> auto rally segmentation + highlights, eventually
conversational video-QA, multisport racquet), pricing assumptions, cost-per-video assumptions,
competitive claims.
Then WEB-RESEARCH (cite URLs for every claim): (1) competitive landscape 2025-2026 ? SwingVision
(tennis/pickleball), Baseline Vision, Pixellot, Veo, Trace, PlaySight, any BADMINTON-specific
AI/highlight products (search hard: badminton AI highlights app, badminton video analysis India,
shuttle tracking app), pickleball/padel analytics startups, generic AI sports highlight tools
(e.g. Reely, Magnifi, WSC Sports) and whether they serve amateurs; (2) pricing points of those
products (subscription tiers) as willingness-to-pay evidence; (3) India/Bangalore badminton
market signals ? academy counts, badminton participation, sports-tech funding in India; (4) the
'multi-court academy footage' problem ? does anyone solve it; (5) commoditization risk: are
general VLMs (Gemini etc.) already good enough at rally segmentation that the owned-model moat
erodes?
Then CRITIQUE the internal assumptions: which survive contact with the evidence, which don't,
what's missing entirely (distribution, capture hardware, who pays ? player vs coach vs academy).
End with an honest pmf_verdict ? not cheerleading: where is the strongest wedge, what would
falsify it fastest, what should be validated before more engineering.
```

2. user (2026-06-10T05:33:20.843Z)

No files found

3. user (2026-06-10T05:33:21.750Z)

```
# PMF & Market Study ? AI Highlight-Reel Tool for Self-Recorded Racket Sports

> **Research snapshot ? 2026-06-07.** Figures are cited as-of with source + year;
> validate live before relying on them for investment/pricing decisions. This is the
> market-research-of-record behind the PMF section of the living
> [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) tracker (§6). Product framing:
> a closed-source hosted API/SaaS (plus local mode) that ingests long-form,
> self-recorded, **static-camera, full-court amateur** racket-sports video and
> auto-detects rallies and compiles highlight reels. Primary sport: badminton;
> sport-generic architecture roadmapped to tennis, pickleball, table tennis, cricket.

> **Confidence convention.** Figures from named industry reports, federations, or
> company pricing pages are tagged with source + year. Inferred items are tagged
> **(estimate)** / **(low confidence)**. Emerging-category market sizes vary widely
> between vendors; ranges are shown rather than false precision.

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## 1. Executive summary

The product sits at the intersection of three favorable trends: (a) a fast-growing
**sports analytics market** (~USD 5?6B in 2025, ~18?22% CAGR to USD 24?36B by ~2034
depending on vendor); (b) a structural shift to **amateur self-recording** on action
cameras and phones (action-camera market ~USD 6.5B in 2024, sports the largest
segment); and (c) **strong racket-sport participation** ? 220M+ badminton players,
106M tennis players (2024), and a near-vertical pickleball curve (~19.8M US players in
2024, +45.8% YoY).

The central tension is the **static-camera constraint**: the tool works best on fixed,
full-court amateur footage and *fails* on broadcast/panning footage. This is both a
moat (most competitors target team sports / broadcast) and a filter on the addressable
user ? the customer must already set a camera on a tripod behind the court. That
behavior is concentrated in **coaches/academies, clubs, and serious enthusiasts**, not
the median casual player.

**Recommended beachhead (strongest single):** **badminton coaching academies and
serious club/competitive players in India** ? a large, rapidly formalizing academy
ecosystem (Khelo India tailwind), genuine fixed-angle recording behavior, existing
willingness to pay for coaching, an underserved tooling gap (incumbents are
team-sports/broadcast), and a direct-but-immature competitor field. The product is
*native badminton-first* ? a genuine edge here. Secondary: **(2) US/global pickleball
players & clubs** (capital-rich, fastest-growing, but SwingVision already strong);
**(3) badminton academies/clubs in East/SE Asia + Denmark** (highest density).

**Pricing** clusters into hardware-bundled auto-filming (Veo, Trace, Pixellot, Spiideo)
and software-only phone AI (SwingVision USD 179.99/yr; BadPro+ pay-per-use). A
software-only BYO-camera model undercuts the hardware players; realistic anchors
**(estimate): B2C USD 60?180/yr; B2B academy/club USD 300?1,500/yr per location.**

**Top risks:** (1) amateurs won't reliably set up a tripod ? TAM collapses to
coaches/clubs; (2) free/near-free alternatives cap B2C WTP; (3) capture-quality
variance is the core failure mode; (4) badminton's strongest markets (India, Indonesia,
China) are the most price-sensitive and hardest to bill at Western SaaS prices.

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## 2. Market sizing & trends

### 2.1 Sports analytics / sports-video market
- **2025 size:** ~USD 5.47B (Precedence Research) to ~USD 5.79B (Fortune Business
  Insights); other vendors USD 5.6?6.0B. *(2025)*
```

- **Growth:** CAGR 15.7%?22.5%. Precedence ? ~USD 29.75B by 2034 (20.63%); Fortune BI ? USD 24.03B by 2032 (22.5%); MarketsandMarkets (narrower) USD 4.75B by 2030 (15.7%); Future Market Insights ? ~USD 36.2B by 2035 (22.1%). *(reports 2024?2025)*
- **Video analytics is the leading segment** within sports analytics (Grand View / Fortune BI) ? the directly relevant sub-market.
- **Interpretation (estimate):** the *amateur/club auto-highlight* slice is a small but fast-growing fraction; most analytics revenue today is pro/broadcast/team-sports. The product targets the long tail the big numbers under-serve.

2.2 Action-camera / amateur recording adoption
- **2024 size:** ~USD 6.5B (GMInsights) ? ~USD 16.9B by 2034 (~10% CAGR). *(2024?2025)*
- **Sports is the largest end-use segment** of action cameras (GMInsights, 2024).
- **GoPro ~47% share, DJI ~20%** (electroIQ/industry, 2024); phones + cheap tripods are the de-facto amateur rig.
- **Implication:** the capture-side behavior the product depends on is already widespread and growing ? *for the segments that record* ? validating "BYO-camera, software-only."

2.3 Participation & popularity by sport and geography

Sport	Participation signal	Source (year)
Badminton	220M+ play regularly; 160+ countries	BWF est. via BadmintonBites (2023)
Badminton ? China	~80M registered, up to ~100M (largest base)	industry compilations (2023)
Badminton ? India	20M+ recreational by 2023; academy boom + Khelo India	industry compilations (2023)
Badminton ? Indonesia	National sport; ~1M club players	industry (est.)
Badminton ? Denmark	100,000+ registered club members (Europe #1)	industry (2023)
Badminton ? Japan/Korea	Major school-club + workplace sport	industry (2023)
Tennis	**106M players (2024)**, +25.6% over 5 yrs (87M in 2021)	ITF Global Tennis Report (2021, 2024)
Tennis ? Asia	~1/3 of players (33M of 87M)	ITF (2021)
Pickleball ? US	**19.8M players (2024)**, +45.8% YoY, +311% over 3 yrs; ~6.2M core	SFIA / Pickleheads / The Dink (2024)
Pickleball ? US courts	~70,000 courts, +55% places to play YoY	SFIA / Pickleheads (2024)
Pickleball ? global	IFP 78 member countries; UK +73% in 2024; ~282M played monthly per PPA claim **(low confidence)**	various (2024?2025)
Table tennis	30M+ players; 227 member associations	ITTF (current)
Cricket	2.5B+ fans; 500M+ in India (mostly *fans*, not self-recorders)	Cricket Rise / Nielsen (2024?2025)

- **Pickleball market value:** ~USD 2.2B (2024) ? USD 9.1B by 2034 at 15.3% CAGR (Market.us/Custom Market Insights, 2024?2025).
- **Badminton equipment market:** ~USD 7.23B global (2023), Asia-Pacific dominant.

Geographic read: badminton density is overwhelmingly Asian + Denmark; tennis is broad/high-income; pickleball is US-dominant, capital-rich, globalizing fast; cricket is huge as *fandom* but its amateur self-recording + static-full-court fit is weak (large field, not a fixed-frame court). For *this product's static-camera fit*, badminton/pickleball/tennis/table-tennis are the core; cricket is a long-tail roadmap item, not a beachhead.

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3. Customer segment evaluation

Segment	Market-size signal	Willingness to pay	Accessibility	Static-camera fit	B2C/B2B	Verdict
Individual amateur/enthusiast	Largest raw TAM (100s of millions)	**Low** ? competes with free; ~USD 5?15/mo ceiling	High via app stores; high CAC/churn	**Mixed** ? most won't set a tripod	B2C	Volume play, weak economics; use as funnel

Coaches & coaching academies	India academy boom; thousands across Asia	**High** ? already sell coaching; USD 300?1,500/yr plausible	Reachable via federations/distributors/social	**Strong** ? already film fixed-angle	B2B/B2B2C	**Best fit** ? buyer, recorder, payer aligned
Clubs & leagues	Denmark 100k+; thousands of clubs Asia/EU	Medium-high; USD 500?3,000/yr	Medium ? committee sales cycle	**Strong**	B2B/B2B2C	Strong secondary; longer cycle
Tournament/event organizers	Many small/regional events	Medium ? per-event/seasonal	Medium ? episodic	**Strong**	B2B	Good per-event upsell; lumpy
Schools & universities	Large (esp. Asia/Japan clubs)	Low-medium ? budget-constrained	Low ? slow procurement	Strong	B2B	Slow; grant/Khelo-India dependent
National/regional federations	Few, high-value	Medium-high; procurement-heavy/political	Low ? long cycles	Strong	B2B	Lighthouse/credibility, not beachhead
Broadcasters / streamers	High value per deal	High	Low ? incumbents entrenched	**Poor ? panning = failure mode**	B2B	**Avoid** ? violates core constraint

Key dynamics: the static-camera constraint is a **B2B filter** ? coaches/academies/ clubs already record from a fixed angle, so the product fits existing behavior (cleanest PMF). B2C is a large low-WTP, high-CAC funnel best used to feed B2B. Broadcasters explicitly violate the constraint ? exclude.

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4. Competitive & pricing benchmarks

Product	Category	Model	Price	Notes (year)
Veo (Cam 3)	Team-sports auto-filming	Hardware + sub	Camera ~USD 1,533 (5G ~1,886) + ~£395?895/yr	15,000+ clubs/80 countries; $115M raised (2025)
Trace	Soccer auto-filming	**Free camera + sub**	USD 180/yr (Basic) ? 300 (Pro)	Camera leased free with sub (2025)
Pixellot (Air)	Auto camera/streaming	Hardware + sub	Camera ~USD 949 + USD 69?167/mo	Institutional/league; 10,000+ clubs (2025)
Spiideo	Auto-filming/analytics	Sub incl. camera	~USD 3,800/yr (1 team) ? 6,100/yr (club)	Institutional budgets (2025)
Hudl	Team video analysis	SaaS tiers	From ~USD 400/yr	Broad team-sports incumbent (2026)
SwingVision	**Tennis/pickleball AI, single iPhone**	Software freemium	**USD 179.99/yr**; free 2 hrs/mo	Closest analog; 200k+ monthly users; $10M+ raised (2025)
BadPro+	**Badminton AI on phone**	**Pay-per-use, no sub**	Per-analysis (? "< one coaching session")	Direct badminton competitor; shot detection, heatmaps, auto-highlights (2025)
Niche pickleball/badminton apps	Various	Mixed	Varies	Fragmented; few true auto-highlight tools (2025)

What buyers actually pay (synthesis): consumer software-only anchors at ~USD 100?200/yr (SwingVision 179.99); hardware-bundled USD 180?900/yr prosumer/club, USD 3,800?6,100/yr institutional; badminton-specific BadPro+ uses *pay-per-use* ? signaling a real badminton AI market but **unproven subscription WTP** (opportunity + warning). **Positioning:** software-only + BYO-camera undercuts hardware players and sells worldwide via SaaS. Realistic anchors **(estimate): B2C USD 60?180/yr; B2B academy/club USD 300?1,500/yr** per location. The differentiators (cheap offline self-improving detection, per-rally stats/tagging, clean licensing) support a **B2B** value story more than a B2C one.

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5. Recommended beachhead

Ranked by **WTP × participation density × existing recording behavior × static-camera fit × competitive whitespace.**

#1 (strongest) ? Badminton coaching academies & serious players, **India**
20M+ recreational players (2023); rapidly **formalizing academy ecosystem** + Khelo India tailwind; academies already record fixed-angle; coaches are buyers who monetize

instruction (high relative WTP); incumbents ignore badminton; the only direct badminton AI (BadPro+) is early with unproven pay-per-use. Product is *native badminton-first*. **Watch-outs:** price sensitivity (need INR-tuned tiers); sell to the **academy**, not the individual.

#2 ? Serious pickleball players & clubs, **US (then UK/Australia/Asia)**
Fastest-growing sport (19.8M US, +45.8% YoY, 2024); capital-rich; ~70,000 courts; strong recording/sharing culture; pickleball is on the roadmap. **Watch-out:** **SwingVision already owns single-camera tennis+pickleball AI** at USD 179.99/yr (200k+ monthly users) ? head-to-head differentiation, not greenfield.

#3 ? Badminton academies/clubs, **East/SE Asia (China, Indonesia, Malaysia, Japan, Korea) + Denmark**
Highest global density (China 80M+; Denmark 100k+ structured club members; Japan/Korea school-club & workplace systems), mature club infra, fixed-court recording. **Watch-outs:** China needs local hosting/partners + is regulatory-complex; SE Asia price-sensitive; Denmark/EU small but high-WTP and easy to pilot.

Bottom line: lead with **India badminton academies** (whitespace + behavior + buyer alignment), use **enthusiast B2C as the funnel**, validate a **club/academy B2B2C motion**, then expand to **pickleball-US** (out-build SwingVision) and **badminton-Asia/Denmark** (density).

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6. Risks & open questions (what would invalidate the thesis)

1. **Tripod/static-camera adoption among amateurs.** If serious amateurs won't set up a fixed camera, B2C TAM collapses to coaches/clubs. *Validate:* observe 3?5 academies + 50?100 enthusiasts on whether they already record fixed-angle.
2. **Free / near-free alternatives** (YouTube, free app tiers, BadPro+ pay-per-use) cap B2C WTP. *Validate:* fake-door pricing at USD 60?180/yr.
3. **Capture-quality variance = the failure mode** (gym lighting, off-center angle, occlusion, mixed resolutions). *Validate:* benchmark rally-detection P/R on 100+ real amateur clips; define a "good-capture" spec + auto-reject (ties to `VIDEO_RECORDING_GUIDELINES.md` + the observability reports).
4. **Monetization mismatch in best-participation markets** (India/Indonesia/China are price-sensitive, hard to bill). *Validate:* local pricing/payments pilot; per-academy INR licensing.
5. **Competitive response** (SwingVision or a team-sports incumbent adds racket sports). *Validate:* ship a badminton-quality advantage incumbents can't trivially match (offline self-improving detection, per-rally tagging).
6. **B2B sales-cycle drag** for clubs/federations/schools. *Validate:* single-coach/ academy self-serve before institutional deals.
7. **Clean-licensing positioning** matters mainly to B2B buyers ? confirm it's a real purchasing criterion via 5?10 buyer interviews.

Cheapest next steps: (a) 10?15 discovery interviews with Indian badminton academy owners/coaches; (b) a capture-quality benchmark on real amateur footage; (c) a fake-door pricing test across B2C (~USD 99/yr) and B2B academy (~USD 600/yr).

---

7. Sources

- Sports analytics market: Precedence Research; Fortune Business Insights; Grand View Research; MarketsandMarkets; Future Market Insights. *(2024?2025)*
  - https://www.precedenceresearch.com/sports-analytics-market
  - https://www.fortunebusinessinsights.com/sports-analytics-market-102217
  - https://www.grandviewresearch.com/industry-analysis/sports-analytics-market
  - https://www.marketsandmarkets.com/PressReleases/sports-analytics.asp
  - https://www.futuremarketinsights.com/reports/sports-analytics-market
- Action camera market: https://www.gminsights.com/industry-analysis/action-camera-market (2024); https://www.precedenceresearch.com/action-camera-market (2024);

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  https://electroiq.com/stats/action-camera-statistics/ (2024)
- Badminton participation/market: https://badmintonbites.com/how-many-badminton-players-are-there-in-the-world/ ;
  https://stringsandpaddles.com/where-is-badminton-played-10-countries-where-its-popular/ ;
  https://en.wikipedia.org/wiki/Badminton ;
  https://www.marketgrowthreports.com/market-reports/badminton-market-110049 ;
  https://www.thesportsschool.com/top-10-badminton-academies-in-india/
- Tennis: https://www.itftennis.com/en/news-and-media/articles/itf-global-tennis-report-participation-hits-106-million-in-five-years/ ;
  https://sgbonline.com/itf-global-tennis-report-2021-shows-worldwide-participation-gains/
- Pickleball: https://sfia.org/resources/as-pickleball-continues-unprecedented-growth-in-every-age-group-and-region-for-third-straight-year-significant-investments-still-needed-for-court-and-facility-demand/ ;
  https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-nearly-20-million-in-2024/ ;
  https://www.pickleheads.com/guides/pickleball-statistics ;
  https://www.custommarketinsights.com/report/pickleball-market/ ; https://market.us/report/pickleball-market/
- Table tennis: https://en.wikipedia.org/wiki/International_Table_Tennis_Federation
- Cricket: https://cricketrise.com/cricket-fans-in-india/ ;
  https://www.nielsen.com/insights/2025/global-cricket-connecting-fans-driving-roi/
- Competitors & pricing: https://www.veo.com/en-us/pricing ; https://traceup.com/pricing ;
  https://www.pixellot.tv/ ; https://www.hudl.com/pricing ; https://swing.vision/subscribe ;
  https://www.badproplusapp.com/
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4. user (2026-06-10T05:33:27.191Z)

```
BACKLOG.md
CODE_MAP.md
COMMERCIALIZATION.md
DEFERRED_HARDENING_PLAN.md
EVALUATION.md
GOLDEN_SET_IMPLEMENTATION.md
GOLDEN_SET_PHASE1_VIDEOS.md
GOLDEN_SET_VENDORING.md
HOW_RALLY_DETECTION_WORKS.md
NEXT_STEPS.md
OWNED_MODEL_IMPLEMENTATION_PLAN.md
PLATFORM_ARCHITECTURE.md
PMF_MARKET_RESEARCH.md
README.md
ROADMAP.md
TRACKNET_WSL_SETUP.md
VIDEO_RECORDING_GUIDELINES.md
archives
```

5. user (2026-06-10T05:33:33.400Z)

```
# Commercialization Tracker

**Status:** ? LIVING ? track this in parallel with product/quality work.
**Owner:** avidullu · **Last refreshed:** 2026-06-07

**Purpose.** Single source of truth for the **commercial** dimension of the project:
licensing/legal posture, codec & output economics, hosting unit-economics, the
golden-set vendoring spend, packaging/distribution, and the product-market-fit
risks that gate monetization. Updated as decisions land ? see the
[Changelog](#changelog) at the bottom.

**Merged from / supersedes:**
```

- `archives/MONETIZATION_AUDIT.md` (2026-05-30 deep-research licensing & economics
  audit) ? kept as the **research-of-record** (per-claim verification transcript,
  primary sources). This doc carries its *current* status forward.
- `archives/COMMERCIAL_READINESS_REVIEW.md` (code-health + PMF assessment) ?
  archived; its commercial findings + the quality-phase priorities live here now.

Premise (unchanged, verified): the product is a **closed-source hosted API** ?
users upload videos to a cloud NVIDIA GPU VM and get highlight reels back. We never
distribute code/binaries, so GPL/LGPL *distribution* copyleft is not triggered ? but
AGPL §13 (network clause), **metered-API terms**, and **video-codec patent
royalties** all still apply.

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1. Commercialization dashboard

Legend: ? done · ? in progress / watch · ? blocker / needs action · ?? needs legal counsel · ? future
Priority: **P0** (do first) · **P1** · **P2** (lowest). Rows are grouped **Pending ? In-flight ? Completed**, each sorted by priority. (C# IDs are stable identifiers ? they don't imply order.)

#	Workstream	Pri	Status	Where it stands	Next action / owner decision
	? PENDING (needs action / not started)				
C5	**Output codec royalties**	**P0**	?	Highlights are still encoded **H.264 + AAC** (`libx264` in `backend/pipeline/stitcher/compiler.py`) ? encode-side AVC patent exposure remains.	Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4/L40)** for HW encode. See §4. Decision: provision Ada-class GPU?
C9	**Packaging / distribution**	**P0**	?	No `pyproject.toml`; the diagnostic `setup.py` shadows the build system.	Design in `DEFERRED_HARDENING_PLAN.md` (#7) ? **awaiting your approval** before code.
C12	**SaaS foundations (async/auth/tenancy)**	**P1**	?	Sync `/api/process` blocks on long videos; no auth/tenancy.	Multi-user/platform strategy in `PLATFORM_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice = `DEFERRED_HARDENING_PLAN.md` (#16).
	? IN-FLIGHT (in progress / watch)				
C4	**Runtime AI strategy (A/B/C + owned model)**	**P0**	?	Live = **(C) Gemini 2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry.	Make "provider registry + paid fallback" a core principle; build the owned model **after** the scaffolding/flywheel are solid. See `PLATFORM_ARCHITECTURE.md` §5A.
C13	**Beachhead validation**	**P0**	?	Market research (`PMF_MARKET_RESEARCH.md`) recommends **India badminton academies** as the #1 beachhead. Unvalidated.	Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R benchmark on 100+ real amateur clips; fake-door pricing (B2C ~$99/yr, B2B academy ~$600/yr). See §6.
C8	**Golden-set vendoring spend**	**P1**	?	ADR-006: **Roora (Bengaluru)** selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors).	Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See `GOLDEN_SET_VENDORING.md`.
C3	**Shuttle-tracking weights provenance**	**P1**	? ??	WASB-SBDT **code is MIT (clear)**; distributed **checkpoints are academic-data-trained** ? not covered by the MIT code grant. MonoTrack/YOLO-NAS banned (noncommercial).	Train-own weights (Phase-4 roadmap) **or** confirm per-checkpoint terms ?? before commercial deploy.
C14	**Owned-model base licensing + training-data policy**	**P1**	? ??	Owned video-QA model (PLATFORM §5A) must build on **Apache-2.0-clean, video-capable** bases ? **Qwen2.5-VL-7B/32B** (Apache; 3B/72B are *not*), **Qwen3-VL-4B/8B**, or **Gemma 4** ? to keep weights private, gate, and monetize. **Reject Gemma 3** (viral derivative clause ? permanent Google PUP), Llama Vision (naming + 700M-MAU), `gpt-oss` (text-only). **Hard rule: never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs** (competing-model clause). The moat is the **consented dataset + eval harness**, not the model.	Lock to Apache-2.0; **open-model teachers only**; separate explicit **training opt-in**; **exclude minors** from the training pool; per-clip consent ledger. See PLATFORM §5A/§7.
C6	**Inbound HEVC decode**	**P2**	? ??	Decoding users' GoPro H.264/**HEVC**		

uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain residual; confirm decode-side exposure with counsel before launch. |
C7	**Hosting unit-economics**	**P2**	?	Per-video cost is *cheap* for Gemini (~$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs unmeasured.	**Benchmark $/video & latency on the target VM** for A/B/C before pricing. See §5.
C10	**Value layer (product surface)**	**P2**	?	Minimal exports shipped (EDL/JSON/CSV + `basic_stats`, PR #51).	Deepen: per-rally tags/notes that feed golden, per-player/match stats, clip gallery. See §6.
C11	**Input-quality enforcement**	**P2**	?	Per-video observability reports surface a customer verdict + footgun flags incl. audio-present (PR #50).	Turn more `VIDEO_RECORDING_GUIDELINES` dimensions into validators/report rules.
C2	**`supervision` ByteTrack pin**	**P2**	?	Pinned `<0.30.0` (deprecation ahead).	Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one.
	? COMPLETED				
C1	**AGPL / copyleft removal**	?	?	ADR-005 shipped: AGPL `ultralytics` YOLOv8 replaced with **torchvision Faster R-CNN (BSD) + `supervision` ByteTrack (MIT)**. No AGPL code/weights in the stack.	Done. Keep new deps permissive.

---

2. Licensing & dependency posture (refreshed risk table)

Verdicts updated against the current stack (`requirements.txt` + code). Original 2026-05-30 verdicts and the 25-claim verification transcript are in `archives/MONETIZATION_AUDIT.md`.

Dependency	License	SaaS obligation?	Current verdict	Notes
~~`ultralytics` + `yolov8n.pt`~~	AGPL-3.0	Yes (§13)	? **REMOVED**	Replaced per ADR-005 ? no longer in the stack.
torchvision detectors (Faster R-CNN/RetinaNet)	BSD-3	No	? LIVE	The chosen replacement; person = COCO class (see `yolo_hybrid.py` `_PERSON_CLASS_ID`).
`supervision` (ByteTrack)	MIT	No	? LIVE	Pinned `<0.30.0` ? migrate before bump (C2).
WASB-SBDT (shuttle)	MIT (code)	No	? code clear	Weights provenance open (C3).
MonoTrack / YOLO-NAS	noncommercial	?	? BANNED	Do not use.
Gemini API (paid tier)	service terms	Paid: no training/review	? LIVE WATCH	Must bill the paid tier to keep user video private (C4).
H.264 / HEVC / AAC codecs	patent pools	royalties on encode/decode	? ??	Separate from ffmpeg's license. Encode-side open (C5); decode-side residual (C6).
ffmpeg/ffprobe, google-genai, torch, opencv, numpy, fastapi/uvicorn/pydantic/jinja2/dotenv	LGPL/Apache/BSD/MIT	No	? CLEAR	Invoked as subprocess / imported; all permissive.
`obsreport` (reports)	permissive (private)	No	? soft dep	Pin the `git+ssh@vX` tag once the repo is published (BACKLOG).

Bottom line: the two hard blockers from the audit (Ultralytics; banned weights) are resolved/avoided. Remaining commercial-legal items are **codec royalties (C5/C6)** and **weights provenance (C3)**, plus the metered-API privacy posture (C4).

---

3. Runtime AI strategy ? A/B/C

The semantic rally-boundary step is the one external/metered dependency. Three paths (from the audit), with current standing:

- **(A) Eliminate ? offline learned segmenter.** ActionFormer (MIT) / our own learned classifier on `candidate_segments.stats`. **Best steady-state margin** (no per-call cost, no external dep) and matches the Phase-4 roadmap (substrate = WASB shuttle-motion, decided in ADR-004). **Most engineering.** ? *target*
- **(B) Self-host VLM.** Qwen2.5-VL-7B/32B (Apache-2.0) or InternVL2.5-8B (MIT). No per-call fees; pays in GPU VRAM (~16?24 GB for 7B). ?? Qwen 3B is noncommercial.

- **(C) Keep Gemini 2.5 Flash (paid).** Fastest, cheap per video, video-native;
  ongoing metered cost + external dependency. **? current live path.**

Hosted alternatives for (C) comparison: Anthropic Claude (no training on
inputs/outputs; customer owns outputs; **no native video** ? weaker here);
OpenAI (no training by default; ZDR available); Gemini remains strongest video-native.

---

4. Codec / output economics (the live gap ? C5)

The trap: ffmpeg being free (LGPL) does **not** grant the *patent* rights to the
codecs it implements. Royalties attach to the act of encode/decode.

- **Encode side (we control it):** highlights currently emit **H.264/AAC**. AVC (Via LA)
  is first 100k units/yr royalty-free with a permanent "free to end-users over the
  internet" provision ? a small SaaS likely falls under thresholds, but ?? confirm.
- **Mitigation:** emit **AV1 (SVT-AV1, BSD + AOMedia royalty-free patent grant) + Opus
  (royalty-free)**. NVENC AV1 HW-encode needs **Ada-class GPUs (L4/L40)**; Ampere/Turing
  must use **SVT-AV1 (CPU)**.
- **Decode side (unavoidable, C6):** inbound GoPro **HEVC** decode ? fragmented pools,
  highest uncertainty; low residual.

Recommendation: provision an **Ada-class GPU (L4/L40)** and standardize highlight
output on **AV1 + Opus** ? removes the encode-side royalty question almost entirely.
Implementation lives in `backend/pipeline/stitcher/compiler.py` (codec args).

---

5. Hosting unit-economics (C7)

Reference workload: ~30-min 1080p match on a cloud NVIDIA GPU VM.

- **(C) Gemini 2.5 Flash (paid):** input $0.30/1M tokens, output $2.50/1M; video ~263
  tok/s ? 10-min clip ? 158k input tokens ? **~$0.05 input**. Cheap per unit; the
  economic risk is **volume × latency**. Flash-Lite is cheaper; Pro ~10×.
- **(B) Self-host Qwen2.5-VL-7B:** $0 API; cost = GPU-hours / throughput; needs ?24 GB
  VRAM (L4/A10). $/video vs Gemini is **open ? measure**.
- **(A) Offline ActionFormer/learned:** cheapest at inference; cost = one-time training
  + light GPU. **Best steady-state margin.**

> The A/B/C $/video comparison depends on GPU instance + batch throughput ?
> **benchmark on the target VM before pricing the product.**

---

6. Product-market fit & the quality/data phase

Beachhead (from market research, 2026-06-07 ? see [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md))
The static-camera constraint is a **B2B filter**: the buyer must already record from a
fixed angle, which points away from the median casual player and toward coaches/clubs.
Ranked beachheads:
1. **Badminton coaching academies & serious players, India** *(strongest)* ? 20M+ players,
   formalizing academy ecosystem (Khelo India), genuine fixed-angle recording, coaches
   who already monetize, badminton whitespace vs incumbents. Sell to the **academy**, not
   the individual; needs INR-tuned pricing.
2. **US/global pickleball players & clubs** ? fastest-growing sport (19.8M US, +45.8% YoY)
   but **SwingVision already owns single-camera tennis/pickleball AI** (USD 179.99/yr) ?
   head-to-head, not greenfield.
3. **Badminton academies/clubs in East/SE Asia + Denmark** ? highest density; China needs

local hosting/partners; Denmark/EU small but high-WTP, easy to pilot.

Pricing anchors *(estimate)*: software-only BYO-camera **B2C USD 60?180/yr**, **B2B academy/club USD 300?1,500/yr** per location (undercuts hardware-bundled Veo/Pixellot/ Spiideo; comparable to SwingVision software-only).

PMF risks (from the readiness review)
- **Capture quality is the gate.** Accuracy is capture-conditional; broadcast/handheld footage is a failure mode. The wedge (static, tripod, full-court) is defensible but narrows the addressable market ? enforce/guide it (C11, `VIDEO_RECORDING_GUIDELINES`).
- **Value beyond raw segments is table stakes.** Time-saved editing alone is weak; per-rally tags, stats, and a correction loop are what make it sticky (C10).
- **Golden labels are the recurring blocker** across the offline segmenter, audio fusion, and measurable quality ? the highest-leverage investment (C8).

Refined priority order (quality/data phase ? "focus on quality once new videos land")
Carried from the readiness review; commercialization items above are tracked in parallel.

1. **Golden-set expansion + data-flywheel UX** (highest with new labeled videos): ingest the new batch, expand calibration/CV/regression baselines, make label contribution + safe retraining easy and non-contaminating (eval/training separation).
2. **Hybrid unification + observability hooks** (deferred item 7, now higher): extract shared logic from `tracknet_hybrid`/`wasb_hybrid` before heavy AI timing/metrics land.
3. **Detector/windowing quality iteration** with new data: re-tune velocity/merge/ min-duration + `rally_gate`; validate WASB on real video; trial audio fusion (ADR-007).
4. **Deeper value layer** (beyond minimal exports, C10): per-rally tagging/notes ? `candidate_segments`, per-player/match stats, clip gallery, human-correction loop.
5. **Input-quality enforcement & guidance polish** (C11): more guideline dimensions ? validators/report rules; "your video is good enough" UX.
6. **Multi-sport maturation + recording best practices.**
7. **SaaS / multi-user / async foundations** (C12) ? lower until monetization is firm.
8. **Polish & distribution** (C9): installer story, first-run path, codec/output, economics.

Recommendation: with the mid-week labeled-video batch, spend ~80% on #1 + #3 (flywheel + quality iteration on real data), then unification (#2) and deeper value (#4).

---

7. Open questions needing owner / counsel sign-off ??

1. **Ultralytics Enterprise price** ? only relevant if a permissive detector ever proves insufficient (fallback). Quote-only.
2. **Per-checkpoint / dataset licenses** for WASB / TrackNetV3 / any VLM ? train-own or confirm each (C3).
3. **Exact Via LA (AVC) / Access Advance (HEVC) rates & free thresholds** ? confirm live figures with counsel before relying on the "small SaaS under threshold" reading (C5/C6).
4. **Real per-video GPU cost & latency** of self-hosted VLM vs Gemini paid at expected throughput (C7).
5. **Provision Ada-class GPU (L4/L40)** for HW AV1 encode? (C5 ? gates the codec switch.)

Changelog

- **2026-06-07** ? Added the market-research-of-record (`PMF_MARKET_RESEARCH.md`): ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13 "beachhead validation" workstream. Wired the summary into §6.

- **2026-06-07** ? Created this living tracker. Merged the 2026-05-30 monetization
  audit + the commercial-readiness review. **Refreshed since the audit:** C1 AGPL
  removal now ? shipped (was the #1 blocker); added C5 as a ? (output is still H.264,
  the audit's AV1 recommendation is not yet implemented); C9 packaging tied to
  `DEFERRED_HARDENING_PLAN.md`; reflected PR #50/#51 (input-quality reports, value-layer
  exports). Archived `COMMERCIAL_READINESS_REVIEW.md`.

See also
- `PMF_MARKET_RESEARCH.md` ? market-study-of-record (sizing, segments, competitors, beachhead; cited).
- `archives/MONETIZATION_AUDIT.md` ? licensing/economics research-of-record (sources + verification transcript).
- `archives/COMMERCIAL_READINESS_REVIEW.md` ? superseded assessment (history).
- `archives/decisions/DECISIONS.md` ? ADR-005 (permissive stack), ADR-006 (vendor process).
- `ROADMAP.md` ? Phase-4 offline segmenter (the C4 "(A) eliminate" path).
- `DEFERRED_HARDENING_PLAN.md` ? packaging (C9), async (C12) designs (pending approval).
- `GOLDEN_SET_VENDORING.md` ? vendor process + Roora pilot (C8).
- `BACKLOG.md` ? `supervision` migration (C2), `obsreport` pin, codec follow-ups.

6. user (2026-06-10T05:33:33.986Z)

Next steps (live) ? owned-model + multisport

Updated: 2026-06-09 (end of the quality-iteration + multisport-recon session). **Authoritative roadmap:**
`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`. **Latest blow-by-blow:** memory `current-state.md` / `session-handoff.md`.
This file = the prioritized "what to do next," refreshed each session.

The recommendation (one paragraph)
Build the **quality-first extraction harness** before anything else: a learned **TAL head** (ActionFormer/ASFormer, MIT)
over trajectory features is the robust path ? it **decouples from the multi-court "background games" confound** that the
unsupervised heuristic (and audio, and spatial court-gating) can't escape (all three tested + failed; it's a *temporal*
rally-vs-non-rally problem). **Framework = ms-swift, base = Qwen3-VL** (Gemma-4 AMBER/bench) come **later** for the
conversational/grounding upgrade ? start with the head. **On-device is deferred** (export-clean ops now so it stays
reachable). Everything stays **MIT/Apache-2.0/BSD** (code + data + weights). Bottleneck = **data + per-sport trajectory
detectors**, not model architecture.

Latest findings (2026-06-09)
- **6-match badminton LOVO:** learned prototype **0.583** vs heuristic **0.480** (learned leads since n=3, AUC 0.83?0.92,
  decoupled from multi-court). The "contrast" metric (in-rally÷dead-time visibility) predicts heuristic F1 monotonically.
- **?? Multisport (Roora pickleball + TT):** labels **ingest cleanly** (`curate convert-cvat` ? 6 rallies/sport, sane).
  WASB-transfer SURPRISE (directional, n=6 each ? TINY): the badminton shuttle detector segments **TABLE TENNIS at
  F1 0.909** out-of-the-box (!) but **pickleball only 0.308**. ? **The per-sport-detector blocker is SPORT-DEPENDENT:** TT may
  be servable with WASB short-term; pickleball needs a proper (MIT/Apache) ball detector. Confirm/deny with more videos.
- **Owner field notes (2026-06-09):** camera height/distance **varied on purpose** (great generalization data); **multi-court
  adjacent-court SOUND dominates the signal** ? empirically reconfirms audio-foreground isolation won't work + the

robustness-first learned-model path. Multi-court is the realistic (academy) environment, so this *is* the target distribution.

Prioritized next steps
1. **[owner greenlight] Phase-0 (M0.x), $0/local:**
   - **M0.4 ? Gen-0 TAL harness** (first-class): productionize `rally_seq_proto.py` ? one-command train?eval?**ship-gate** over
     ActionFormer/ASFormer, reusing `metrics.match_intervals`. **Define the ship-gate target** (explicit F1@tIoU; floor = beat
     heuristic LOVO 0.480) ? "very high P/R" must be operationalized, not just "beat the floor."
   - **M0.1 ? corpus spine + the per-video event-index contract** (the conversational seam: extract ? event store ? text-to-SQL).
   - **M0.3 ? compliance:** purge Gemini-derived training labels (`training.label_candidates`); WASB **C3** own-weights/clean-swap (gates ship).
2. **Multisport thread:**
   - Ingest the Roora pickleball/TT rally CSVs into the corpus (collector `import-local`).
   - **Get more pickleball + TT videos** ? re-run the WASB-transfer eval to confirm/deny TT?0.91, pickleball?0.31 (n=6 is too few).
   - Once each sport has ?3 matches, run a **merged prototype LOVO across badminton + TT (+ pickleball)** ? the "how would a
     merged model do" experiment. Identify a clean MIT/Apache ball detector for sports where WASB doesn't transfer (pickleball).
3. **Housekeeping:** **rename repo off "badminton"** (multisport); confirm collector = corpus/event-store system-of-record.

Pointers / artifacts
- Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`
  (2026-06-09 banner) · Quality: `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
- Eval corpus: `output/human_lovo_manifest.json` (badminton n=6). Roora CSVs: `%TEMP%\roora\{pickleball,tabletennis}_rallies.csv`;
  WASB-transfer trajectories: `%TEMP%\{pickleball,tabletennis}_traj.csv`. Code: `backend/eval/{eval_partial_labels,rally_seq_proto,calibrate_local}.py`.
- ? No owned-model implementation until owner greenlights a milestone. Committed next PLATFORM slice = async job model (single-tenant).

7. user (2026-06-10T05:33:38.410Z)

Roadmap ? from annotated data to a self-improving, offline rally segmenter

This document is the **narrative thread**: how the indexer evolves from an
AI-API-dependent prototype into a self-contained tool that learns rally detection
from its own commercially-licensed, annotated video corpus. It explains *how we
got here*, *why each step matters*, and *how it shapes the final tool*. Decisions
are recorded as ADRs in DECISIONS.md; the scoring mechanics live
in EVALUATION.md.

The thesis

The indexer detects badminton rallies in long recordings. Its strongest mode
(`yolo_hybrid`) is a two-stage pipeline: cheap **permissive person-detection
player-motion** filtering (Stage 1, local ? torchvision Faster R-CNN + ByteTrack
tracking) followed by **Gemini** semantic refinement (Stage 2, paid API). That
works, but it costs money, needs the network, and never improves on its own.

> *Naming note:* the `yolo_hybrid` registry key is retained for back-compat, but
> AGPL YOLOv8 was removed and replaced with a BSD/MIT-licensed torchvision +
> ByteTrack stack ? see **ADR-005**. "YOLO" in this doc means the local
> player-motion gating stage, not the Ultralytics library.

A sibling project, **sports-data-collector**, curates sports videos with per-rally `[start, end)` annotations and a commercial-license flag. The thesis of this roadmap: **use that annotated, commercially-usable data to (1) measure the pipeline objectively and (2) train an offline replacement for the paid Gemini stage** ? turning the tool into one that improves as more data is curated, runs air-gapped, and generalizes across sports.

Why it matters: it removes the per-run API cost and network dependency, gives us a defensible quality metric instead of eyeballing, and creates a flywheel ? more curated matches ? a better offline segmenter ? better highlights.

The four phases

Phase 1 ? Bridge the repos (DONE)
A sport-agnostic `curate export` in the collector emits the indexer's `start,end` ground-truth CSVs plus a `manifest.json` pairing each CSV with its video, gated on the `is_commercial` flag. ? *Why:* one clean producer/consumer contract that works for all five supported sports (badminton/tennis/cricket/pickleball/table-tennis) alike. See **ADR-002**.

Phase 2 ? Acquire full-resolution video (DONE)
The corpus was 256×144 (the original crawl used `worstvideo`). The `fetch` command re-acquires every video at **best available resolution (1080p here)**, **in place** (preserving annotations), via **authenticated Firefox Premium cookies** and yt-dlp's `default` client. ? *Why:* YOLO/shuttle detection needs real pixels; and the journey surfaced non-obvious traps (SABR streaming ? 144p; Chrome DPAPI; rate limiting) now encoded in the tooling. **22/22 fetchable matches are now 1080p.** See **ADR-003**.

Phase 3 ? Evaluate + tune (IN PROGRESS)
`run_phase3_eval.py` walks the manifest, processes each video, and scores detected rallies against the ground truth ? per video and corpus-aggregate (precision/recall/ F1, mean IoU, boundary error), for both the `candidates` (raw detector) and `final` (refined) stages. ? *Why:* the candidates-vs-final split tells us *where* error lives ? a **detection** problem (candidates miss rallies) vs. a **segmentation** problem (candidates catch them, refinement is loose). That diagnosis decides where effort goes, and establishes the baseline any trained model must beat.

Phase 4 ? Learn an offline, API-free segmenter (NEXT)
Replace the paid Gemini refinement with a **lightweight classifier trained on the rally annotations**, consuming features the pipeline already computes and stores (`candidate_segments.stats`). No external API, no new labeling. ? *Why:* this is the payoff ? a free, offline, self-improving rally mode. The **substrate was chosen from data**: the ADR-004 update (2026-06-02) measured the YOLO/player-motion candidate stage at **recall ~0.05 @ IoU 0.5** on `shuttleset_8` (median best coverage of a rally only ~20%) ? too low a ceiling to build on. So the offline segmenter's substrate is **WASB shuttle-motion** (ADR-001), not player-motion. See **ADR-004**.

Complementary local cues (beyond the segmenter swap)

Once the substrate is WASB shuttle-motion, the next lever is **fusing additional fully-local cues** to raise rally recall/precision without reintroducing a paid API. The first one explored is **audio "thwack" hit-detection** (**ADR-007**, accepted as a direction 2026-06-07): a pure-numpy spectral-flux onset detector in `backend/pipeline/audio/`. The **E1 GO/NO-GO probe** has run on real GoPro audio ? verdict **AMBER / leaning-negative**: recall signal is real (hit-clusters recover WASB-missed rallies) but the classical-onset precision ceiling (~2.2×) is too weak to fuse cleanly, and a band-pass doesn't rescue it. The only path past that is a *learned timbre classifier* ? which, like Phase 4, is **gated on golden labels**. See `AUDIO_E1_FINDINGS.md`. Pose/court serve-gating is parked as the #2 cue.

The recurring blocker across Phases 3?4 and the audio work is the same: **more golden labels** is the single highest-leverage move (see the cross-cut note below).

How this shapes the final tool

```
89      - **Offline & free by default.** The production rally mode becomes a local model;
90        Gemini is kept only as an upper-bound reference, not a dependency.
91      - **Self-improving.** Every newly curated, annotated match is more training signal;
92        re-training is a local, deterministic step.
93      - **Measurable.** Every change is scored against the golden set, so "better" is a
94        number, not a vibe.
95      - **Sport-generic.** The label ? feature ? classifier machinery and the export
96        bridge already span sports; adding a sport is data, not new architecture.
97      - **Commercially clean.** Only `is_commercial` data flows through the bridge, so the
98        trained model and any derived product rest on a licensable foundation.
99
100     ## Status at a glance
101
102     | Phase | What | Status |
103     |------|------|--------|
104     | 1 | Export bridge (`curate export` ? manifest + CSVs) | ? Done |
105     | 2 | 1080p acquisition (`fetch`, in-place, cookies) | ? Done (22/22) |
106     | 3 | Eval + tuning loop (`run_phase3_eval.py`) | ? In progress |
107     | 4 | Offline learned segmenter (WASB shuttle-motion substrate) | ? Next (substrate
decided: WASB, ADR-004) |
108     | + | Audio hit-detection cue (`backend/pipeline/audio/`, ADR-007) | ? E1 probe run ?
AMBER, gated on golden labels |
109
110     > **Cross-cutting:** golden-set ground truth (which powers Phases 3?4 and on-demand
regression testing)
111     > is produced via a vendor process + pluggable annotation provider ? see
112     > [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md) and **ADR-006**. As of 2026-06-02
the annotation
113     > guideline is authored (collector `docs/annotation/`) and **Roora** (Bengaluru) is
selected for an
114     > initial **3-sport pilot** (badminton, pickleball, table tennis). Input-capture quality
requirements
115     > that bound achievable precision/recall are tracked in
[VIDEO_RECORDING_GUIDELINES.md](VIDEO_RECORDING_GUIDELINES.md).
116     >
117     > **Observability:** every processed video now emits an always-on per-video diagnostics
118     > report (spec-driven warnings/errors with FAQ refs, soft-dependency on `obsreport`), so
119     > pipeline regressions and footguns surface without re-running eval ? see
120     > [CODE_MAP.md](CODE_MAP.md) §2 (reporting/observability).
121
```

8. user (2026-06-10T05:33:43.449Z)

```
1:# Platform Architecture ? Multi-User "Assistant Coach"
2:
3:**Status:** ? Strategy / research ? for review. No code yet.
4:**Date:** 2026-06-07 · **Owner:** avidullu
5:
6:**Purpose.** Map the path from today's **single-user, local** tool to a **multi-tenant
7:"assistant coach" platform**: users upload videos, get highlights **stored and
8:browsable by history**, and choose **where their data lives** (Google Drive / cloud
9:bucket / local) and **where the compute runs** (our GPU / their VM). The organizing
10:principle is to **decouple the layers** ? identity, ingestion, storage, orchestration,
11:execution/compute, the detection+highlight pipeline, and data-serving ? behind small
12:**pluggable interfaces**, reusing the registry pattern the codebase already uses for
13:segmenters/providers/validators/sports/annotations.
14:
15:**Product framing (the north star *and* the v1 line).** This is an **assistant coach**,
16:not just a clipper ? but the strategy is **phased and unambiguous about scope**:
17:- **v1 (concrete, near-term):** **rally detection ? highlight-reel compilation ? per-user
18:  history ? a coach/user correction loop that feeds the golden set** ("assistant-coach
19:  *lite*"). Essentially today's pipeline made multi-user ? this is the real v1 deliverable.
20:- **Mid-term:** **rich rally/shot semantic metadata** (shot type, rally outcome) and the
21:  **owned video-QA model** (§5A; §6 P4?P5).
22:- **North-star vision (strictly future ? *not* v1):** **gait / biomechanics, per-athlete
```

23: identifying templates, and deep coaching feedback under a supervising coach** (§6 P6).
24: Inspiring, but explicitly deferred ? and the part that raises the **biometric** privacy
25: bar (§7).
26:
27:The buyer is the **academy/coach** (beachhead in
28:[`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md)); **power users self-serve**. Two
29:consequences ripple through the design: (1) the data model keeps an **optional coach ?
30:athlete overlay over a generic core** (§4.1) ? not baked in; (2) gait/biometrics is
31:**special-category data**, gated and deferred (§7).
32:
33:> **Competitive note (validated by research):** none of Veo / Hudl / SwingVision /
34:> Spiideo offer true BYO-storage or BYO-compute ? they're vendor-cloud SaaS
35:> (SwingVision uniquely runs *edge* compute on the user's iPhone). So **"your video,
36:> your storage, your compute"** is a genuine differentiator for privacy-sensitive
37:> academies ? and SwingVision proves user-side compute is commercially viable. All four
38:> share the same **teams ? athletes ? clips** data model, which we adopt (§4).
39:
40:> Scope: this is a **direction-setting doc**. It does not start implementation; the
41:> first infra slice (an async job model) is the on-hold #16 in
42:> [`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md). Commercial constraints come
43:> from [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) (C4 paid-LLM no-train; C12 SaaS
44:> foundations) and `archives/MONETIZATION_AUDIT.md`.
45:
46:## 1. Current state & gaps (what we're evolving from)
47:
48:| Concern | Today | Gap for multi-user |
49:|---|---|---|
50:| Identity / tenancy | none | no users, orgs, auth, or isolation |
51:| Storage | local `filepath` string; output to `output_dir` | no abstraction; can't BYO
Drive/bucket; raw video must be on the box |
52:| Data model | `videos` keyed by file **MD5**, `play_segments`, `events`,
`candidate_segments` (SQLite) | no owner/athlete link; **highlights aren't a first-class
persisted entity** (compile writes a file and returns a path); no per-user history |
53:| Execution | pipeline runs **in-process, synchronously** in `/api/process` | blocks; can't
dispatch to a user's VM/GPU; no job status |
54:| Pipeline | **already pluggable** (segmenter/provider/validator/sport/annotation registries)
| reuse as-is ? this layer is the model for the rest |
55:| API | `/api/process`, `/api/query`, `/api/compile`, `/api/config`; no auth | needs auth,
per-tenant scoping, async job + history endpoints |
56:
57:Grounding: `backend/infrastructure/database.py` (schema + `PRAGMA user_version`
58:migration ladder, v1), `backend/main.py` (endpoints, sync flow), the registry pattern
59:in `backend/pipeline/segmenters/base.py::SegmenterRegistry` and
60:`backend/providers/base.py`.

9. user (2026-06-10T05:33:51.064Z)

159: agents + health heartbeats + clear worker status in the UI**, and budget the support cost.
203:| Model | Isolation | Ops cost | Best for |
228: writes artifacts + DB rows. **Observability reports** (already per-video) become the
325: cost pain ? below that, the paid API is cheaper than a standing GPU and the second stack
451:    optional global-adapter + per-user-adapter stacking; per-user training cost at scale.
541:**base-model / teacher-distillation licence violations**; cost of hosted GPU (training
563:**Owned model & licensing (§5A):** [Qwen2.5-VL](https://qwenlm.github.io/blog/qwen2.5-vl/) ·
[Qwen2.5 license split](https://qwenlm.github.io/blog/qwen2.5/) ·
[Qwen3-VL](https://github.com/QwenLM/Qwen3-VL) · [Gemma 3
(multimodal)](https://huggingface.co/blog/gemma3) · [Gemma Terms of
Use](https://ai.google.dev/gemma/terms) · [Gemma license-risk
analysis](https://wcr.legal/google-gemma-license-risks/) · [Gemma 4 /
Apache-2.0](https://www.mindstudio.ai/blog/what-is-gemma-4-apache-2-license-commercial-ai-
deployment) · [Llama 3.2 license (700M MAU + naming)](https://www.llama.com/llama3_2/license/) ·
[gpt-oss (text-only, Apache-2.0)](https://openai.com/index/introducing-gpt-oss/) · [Gemini API
terms (compete clause)](https://ai.google.dev/gemini-api/terms) · [OpenAI
terms](https://openai.com/policies/row-terms-of-use/) · [Anthropic: outputs &
training](https://support.claude.com/en/articles/12326764-can-i-use-my-outputs-to-train-an-ai-

model) · [QLoRA practical](https://apxml.com/courses/fine-tuning-adapting-large-language-models/chapter-4-parameter-efficient-fine-tuning/practice-qlora-fine-tuning) · [VLM LoRA cookbook](https://huggingface.co/learn/cookbook/en/fine_tuning_vlm_trl) · [GDPR Art. 9](https://gdpr-info.eu/art-9-gdpr/) · [athletes & biometric data](https://www.architectureandgovernance.com/applications-technology/biometric-data-and-athletes-privacy-law-and-compliance-implications/) · [federated learning + DP](https://research.google/blog/federated-learning-with-formal-differential-privacy-guarantees/) · [self-hosted LLM VRAM/cost](https://www.gpu-mart.com/guides/self-hosted-llm) · [Gemini API pricing](https://ai.google.dev/gemini-api/docs/pricing)

10. user (2026-06-10T05:33:52.007Z)

9:Every deferred piece (conversational-QA, on-device, multisport, the VLM) must be **architecturally envisioned
21:- **The moat = the consented dataset + the eval harness, not the weights.**
30:- ? **Base-model pretraining provenance is unauditable for *every* open VLM (incl. Qwen3-VL).** Our "no paid-LLM
32: risk-acceptance** is required to use any open VLM. (Governed by the base's license, not our pipeline.)
40:- **Base (when we reach the VLM): Qwen3-VL (Apache-2.0, uniform across sizes)** ? long-video context, purpose-built
46:- **START WITH THE TAL HEAD, NOT THE VLM** (the decisive call): fine-tuned VLMs **miss strict temporal boundaries**
47: (best RL-post-trained video VLM ? R@0.7 50% on open-domain benchmarks; "coarse regions, not accurate boundaries").
50: and **doubles as a pseudo-label teacher + the honest baseline the VLM must beat.** So: **head now ? VLM later**,
51: strictly sequenced. When the VLM comes, prefer **two-stage** (VLM coarse-proposes, head refines boundaries) over
52: wholesale replacement. *(Open: the VLM-ceiling number is open-domain; our amateur footage may differ ? re-measure.)*
56:over a weak extractor exposes the weakness). **Reject end-to-end-VLM-answers-each-turn** (expensive; bad at exact
58:- **Architecture = "extract once, query many":** EXTRACTION (TAL head ? later VLM) emits structured timestamped
61: VLM job** (SQL counts exactly where VLMs hallucinate).
101:### M0.1 ? The labeled-corpus spine **+ the event-index contract** (the moat) · *greenlight item*
105:labeled_window, derived_clip_ref, confidence}`. 3 transcoders (heuristic/LogReg · TAL ActivityNet-JSON · VLM clip?QA),
139:- **Gen-2 (VLM):** ms-swift fine-tune of **Qwen3-VL** (SFT cold-start ? GRPO/RFT with a temporal-IoU-vs-golden reward,
140: our own CoT supervision). **Two-stage** (VLM proposes, head refines). Cloud train on Modal (no-access) / RunPod-Secure.
141:- **Hard pre-serving gates (unmeasured, load-bearing):** the **$/call crossover** (self-hosted VLM vs Gemini) **+ a real
159:trajectory detectors not yet identified · base-model pretraining provenance unauditable (risk-acceptance) · real VLM

11. user (2026-06-10T05:33:58.406Z)

300
301   ### Base model & training (Apache-2.0 only ? licensing is decisive)
302   **Recommended near-term base: `Qwen2.5-VL-7B-Instruct`** ? Apache-2.0, native long-video
303   (dynamic-FPS sampling, temporal mRoPE), mature LoRA tooling. Upgrade path
304   **Qwen3-VL-4B/8B (Apache-2.0)**; **Gemma 4 multimodal (Apache-2.0, 2026)** is a clean
305   Google-stack alternative to evaluate.
306   - ?? **Lock to Apache-2.0-clean bases** so you can fine-tune, **keep weights private,
307   gate behind subscriptions, and never owe open-sourcing or branding**. (Note Qwen2.5-VL
308   is Apache **only at 7B/32B** ? 3B/72B use the custom Qwen license.)
309   - ? **Reject for the flagship:** **Gemma 3** ? custom terms with a **viral derivative
310   clause** (anything fine-tuned on it, *or even trained on its synthetic outputs*, stays
311   permanently bound to Google's *updatable* Prohibited-Use Policy); **Gemma 4's switch to
312   Apache-2.0 removes this**. **Llama 3.2 Vision** ? custom license forces a "Llama-?"

model name + "Built with Llama" badge and a 700M-MAU ceiling. **`gpt-oss`** ? Apache-2.0
but **text-only**, so usable only as an optional text-reasoning layer, never the video
encoder.
- **Training method:** **QLoRA** (? r=64, ?=128) on the 7B ? fits one prosumer GPU and
avoids catastrophic forgetting of general video skills. Build the dataset by converting
the existing rally/shot/segment annotations + golden set into templated **clip?QA
pairs**, expand synthetically with **open-model teachers only** (see constraint below),
and hold out a **golden QA eval** that gates every fine-tune. A few hundred gold pairs ?
5k?20k curated is a realistic band for this narrow domain.
- **Serving economics:** a 7B VLM ? 14 GB VRAM (FP16) / ~3.5 GB (INT4); a ?16 GB GPU is a
comfortable serve target (vLLM for throughput). Self-hosting wins **only** once you have
sustained video-QA volume **+** a defensible consented dataset **+** real per-call API
cost pain ? below that, the paid API is cheaper than a standing GPU and the second stack
is overhead. Hence: **paid-primary now ? owned-primary / paid-fallback later.**

Base-model licensing + training-data policy are tracked as **COMMERCIALIZATION C14**.

Training backend ? managed service vs self-managed (the fine-tune "forge")
Where you run the LoRA/RL fine-tune. Weigh on **video support, weight export/ownership,
smallest servable model, RL support, no-train DPA, and infra burden**.

Decisive finding (researched 2026-06): managed fine-tuning services are
image-based (JSONL + base64 images) ? for video you pre-sample frames and feed them
as image sequences. **Only the self-managed `ms-swift` explicitly advertises *native*
video (+audio) training.** So for true video-QA, a self-managed framework is the
strongest technical fit ? and it also lets you fine-tune the **small 7B** (cheap to
serve) and keep full ownership. Managed services trade some of that for zero infra.

12. assistant (2026-06-10T05:34:09.000Z)

Now I have the internal thesis extracted. Let me run the external web research in parallel batches.

13. user (2026-06-10T05:34:16.754Z)

Web search results for query: "SwingVision pricing 2026 tennis pickleball AI app subscription badminton support"

Links: [{"title":"SwingVision Pro Plans & Pricing | Tennis & Pickleball App","url":"https://swing.vision/subscribe"},{"title":"SwingVision: Tennis Pickleball - App Store - Apple","url":"https://apps.apple.com/us/app/swingvision-tennis-pickleball/id989461317"},{"title":"SwingVision: Tennis Pick... ? Review 2026: Sentiment & Intel","url":"https://marlvel.ai/intel-report/sports/com-mangolytics-swing"},{"title":"SwingVision: AI Stats for Tennis & Pickleball","url":"https://swing.vision/"},{"title":"SwingVision: Pricing, Reviews & Features 2026 | Sports & Fitness Tools","url":"https://sports.toolsinfo.com/tool/swingvision"},{"title":"SwingVision Reviews in 2026","url":"https://sourceforge.net/software/product/SwingVision/"},{"title":"SwingVision: Tennis & Pickleball AI Stats","url":"https://appshunter.io/ios/app/989461317"},{"title":"SwingVision: AI-powered app for automated tennis & pickleball scoring, stats, highlights, and line calling.","url":"https://www.toolify.ai/tool/swingvision"},{"title":"SwingVision: AI Scoring, Stats & Line Calling","url":"https://swing.vision/ambassador-offer"}]

Based on the search results, here's the information about SwingVision pricing and features in 2026:

Pricing

SwingVision Pro costs $14.99 USD per month or $179.99 USD per year, with prices varying by country. SwingVision offers a free trial.

Features

SwingVision Pro provides AI-powered shot tracking, automated scoring, line calling, and detailed match stats on every shot. SwingVision is the #1 tennis & pickleball app for automated scoring, stats, highlights & line calling.

Key features include:
- Automated analysis of tennis or pickleball matches using AI and a single camera, tracking shots, providing real-time line calls, generating automated stats and scoreboards, and creating highlights of your best rallies
- The AI edits out the dead time in between points means you can watch your 2 hour match in 15-20 minutes

Sport Support

SwingVision is designed for tennis and pickleball. The search results did not show any information about badminton support for SwingVision.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

14. user (2026-06-10T05:34:18.781Z)

Web search results for query: "badminton AI highlights app shuttle tracking video analysis 2025 2026"

Links: [{"title":"10 Ways AI Is Being Used in Badminton [+5 Case Studies][2026] - DigitalDefynd Education","url":"https://digitaldefynd.com/IQ/ai-use-in-badminton/"},{"title":"Instant review system for badminton: computer vision system use case - Spyrosoft","url":"https://spyro-soft.com/blog/artificial-intelligence-machine-learning/instant-review-system-for-badminton-computer-vision-use-case"},{"title":"The AI Badminton Trainer enhances players' training and performance - RADL","url":"https://radlwire.com/ai-badminton-trainer/"},{"title":"AI-powered Badminton Video Detection: Enhancing Gameplay Analysis and Training | TechRxiv","url":"https://www.authorea.com/doi/full/10.36227/techrxiv.23708325"},{"title":"Custom AI Badminton Video Analysis Software Development","url":"https://tezeract.ai/industries/badminton-video-analysis-software/"},{"title":"Real-Time Badminton Player Tracking Using Computer Vision","url":"https://www.bacancytechnology.com/blog/player-tracking-with-computer-vision"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://www.mdpi.com/1424-8220/24/13/4372"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC","url":"https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"}]

Based on the web search results, here's what I found about badminton AI highlights apps with shuttle tracking and video analysis capabilities for 2025-2026:

Key AI Technologies and Applications

Real-Time Shuttle Tracking & Analysis

The AI Badminton Trainer features real-time 3D visual analysis that tracks shuttlecock speeds and maps precise trajectories. The system can track shuttlecock speed in real-time up to 300 km/h and map the shuttlecock's trajectory.

AI-powered badminton video detection apps can accurately detect and track players and shuttlecocks in real-time, providing valuable insights into player performance and strategy.

Automated Highlights Generation

Systems generate automatic match highlights based on event detection and action recognition models, recognizing intense moments such as smashes, long rallies, and critical points. This enables players and viewers to automatically generate highlight reels.

Video Analysis Features

AI-powered solutions deliver real-time player tracking, shot classification (smash, drop, clear, net shot), rally breakdowns, and tactical heatmaps, along with shuttle trajectory analysis and court movement patterns.

AI-driven platforms analyze live matches to provide viewers with real-time statistics, graphical insights, and predictive outcomes, making the game more engaging and understandable.

Recent Developments

ITRI's wellness technologies, including the AI Badminton Trainer, were making their U.S. debut at CES 2025, featuring advanced shuttle tracking and strategic gameplay analysis capabilities.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

15. user (2026-06-10T05:34:29.298Z)

Web search results for query: "BadPro+ badminton AI analysis app pricing pay-per-use"

Links: [{"title":"BadPro+ | AI-Powered Visual Intelligence for Badminton","url":"https://www.badproplusapp.com/"},{"title":"BadPro+ - Apps on Google Play","url":"https://play.google.com/store/apps/details?id=com.badproplus&hl=en_US"},{"title":"BadProPlus - App Store - Apple","url":"https://apps.apple.com/us/app/badproplus/id6746172500"},{"title":"GGAB App","url":"https://app.getgoodatbadminton.com/"},{"title":"AI-Powered Desktop App for Badmin Match Performance Analysis","url":"https://www.bacancytechnology.com/case-study/ai-ml/playtech-analytics"},{"title":"AI Software Pricing 2026: 18 Tools Compared (Real Costs)","url":"https://thecrunch.io/ai-software-price/"},{"title":"AI Badminton Analysis app","url":"https://universe.roboflow.com/ai-badminton-analysis-app"},{"title":"Custom AI Badminton Video Analysis Software Development","url":"https://tezeract.ai/industries/badminton-video-analysis-software/"}]

Based on the search results, here's what I found about BadPro+ pricing:

BadPro+ offers AI analysis on demand with no subscription. The app operates on a pay-per-match model, allowing users to pay only for the analyses they use rather than committing to a recurring subscription.

The platform offers multiple tiers:

1. **Individual/Player Level**: Professional analytics for less than one coaching session, with features including real-time shot detection, heat map analysis, and personalized feedback.

2. **Team/Federation Level**: Federation-grade analytics for clubs, federations, and pro teams with roster tools, opponent scouting, and predictive modeling. Pricing is tailored to team size and integrations.

3. **API/Broadcast Level**: Real-time shot data, scoring, and analytics by the call.

BadPro+ was founded by an Olympic badminton athlete and Microsoft engineer, and the app is available on both iOS and Android platforms.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

16. user (2026-06-10T05:34:29.331Z)

Web search results for query: "Baseline Vision tennis AI camera pricing 2026"

Links: [{"title":"Baseline Vision Review | Best Portable Line-Calling System?","url":"https://www.tennisleo.com/baseline-vision-review/"},{"title":"Get Your Baseline Cam Now | Baseline Vision","url":"https://www.baselinevision.com/product"},{"title":"Baseline Vision Tennis Tracking and Line Calling System","url":"https://letsgotennis.com/product/baseline-vision-tennis-tracking-and-line-calling-system/"},{"title":"Baseline Vision - Tennis. Re-invented","url":"https://www.baselinevision.com/"},{"title":"Baseline Vision, The Baseline Cam ? Questfind E-store","url":"https://questfindestore.com/product/baseline-vision-the-baseline-cam/"},{"title":"Baseline Vision ? Portable AI-Powered Tennis Game Analysis Camera System - The SUP Desk (2025)","url":"https://thesupdesk.com/baseline-vision-portable-ai-powered-tennis-game-analysis-camera-system/"},{"title":"Baseline Vision - Enhance tennis skills with AI-driven insights and analytics.","url":"https://oncely.com/products/baseline-vision-discount"},{"title":"Baseline Vision - Products, Competitors, Financials, Employees, Headquarters Locations","url":"https://www.cbinsights.com/company/baseline-vision"},{"title":"Baseline Vision - Motion Sports Australia","url":"https://motionsports.com.au/baseline-vision/"}]

Based on the search results, here's what I found about Baseline Vision tennis AI camera pricing in 2026:

Current Pricing (2026)

The standard price is $2000 USD (or 1800?), with promotional discounts available. For example, a discount code "LEO1" provides $100 off.

Additionally, the standard price is listed at $2299, but there's currently a 30% discount bringing it down to $1999, saving $300.

What's Included

The kit includes the camera, mount, case, backpack, charger, and battery.

Additional Details

- Free worldwide shipping and a 30-day trial period
- A 2-year warranty covering defects in materials and workmanship under normal use

The system is ITF-approved and provides real-time line calling, performance analytics, and AI-powered game analysis for tennis players and coaches.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

17. user (2026-06-10T05:34:40.813Z)

Web search results for query: "pickleball AI video analytics startup PB Vision pricing 2025"

Links: [{"title":"PB Vision - Pickleball analytics powered by AI","url":"https://pb.vision/"},{"title":"PB Vision: Pricing, Free Demo & Features | Software Finder","url":"https://softwarefinder.com/artificial-intelligence/pb-vision"},{"title":"Get Your DUPR Rating with AI & PB Vision - thePicklebase | ThePickleBase","url":"https://www.thepicklebase.com/coaching-tips/future-pickleball-ratings-getting-dupr-rating-power-ai"},{"title":"Pickleball Vision AI - Crunchbase Company Profile & Funding","url":"https://www.crunchbase.com/organization/pickleball-vision-ai"},{"title":"PodPlay x PB Vision: Raising the AI Video Analysis Bar for Pickleball Clubs","url":"https://podplay.app/blog/podplay-x-pb-vision-raising-the-ai-video-analysis-bar-for-pickleball-clubs"},{"title":"PB Vision - App Store - Apple","url":"https://apps.apple.com/us/app/pb-vision/id6467020610"},{"title":"PB Vision API

Partner Guide - PB Vision","url":"https://help.pb.vision/articles/2793895-pb-vision-api-partner-guide"},{"title":"DUPR Blog | DUPR and PB Vision Partner to Shape the Future of AI Pickleball Ratings","url":"https://www.dupr.com/post/dupr-and-pickleball-vision-partner-to-shape-the-future-of-ai-pickleball-ratings"},{"title":"PB Vision - Apps on Google Play","url":"https://play.google.com/store/apps/details?id=com.pb.vision.app"},{"title":"DUPR Blog | How the AI Boom Is Making Pickleball a Success","url":"https://www.dupr.com/post/how-the-ai-boom-is-making-pickleball-a-success"}]

Based on the search results, here's what I found about PB Vision's pricing for 2025:

Consumer Pricing Plans

PB Vision offers a free trial where new users can upload one video (up to 25 minutes) for free to test the analysis.

For paid plans, the Annual Pass is currently priced at $99/year (approximately $8/month), which includes roughly 100 minutes of analysis per month (10-15 games). A Starter Monthly option is available at $19.99/month for players who want to try it out on a month-to-month basis.

API/Partner Pricing

For businesses and partners using the PB Vision API, HD video (1080p or lower) costs $8 per hour of video processed (about $1.60 per 12-minute game), while 4K video costs $12 per hour (about $2.40 per 12-minute game).

Integration with PodPlay

The PB Vision integration is available at no additional cost to PodPlay clubs that have PodPlay Pro monitors, with a usage-based pricing model (per-minute processing) that includes a floor price to cover compute costs.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

18. user (2026-06-10T05:34:43.305Z)

Web search results for query: "India sports tech startup funding 2025 badminton academy technology video analysis"

Links: [{"title":"India?s Sportstech Field: 13 Startups Tapping The Sports Opportunity","url":"https://inc42.com/startups/meet-the-startups-scoring-big-on-the-indian-sports-tech-pitch/"},{"title":"Indian badminton players use artificial intelligence to up their game - Mugglehead Investment Magazine","url":"https://mugglehead.com/indian-badminton-players-use-artificial-intelligence-to-up-their-game/"},{"title":"This startup serves sharp insights and graphics to help ...","url":"https://yourstory.com/2025/03/this-startups-offers-data-driven-insights-and-graphics-help-athletes"},{"title":"??? India?s Sports-Tech Surge: Part 1 ? Data, Performance & the Race for Fan Attention","url":"https://mehttaventuresdubai.substack.com/p/indias-sports-tech-surge-part-1-data"},{"title":"Best Sports Companies to Work for in India 2026 | Wellfound","url":"https://wellfound.com/startups/l/india/sports"},{"title":"Top 7 Badminton startups 2025","url":"https://www.sportstartups.org/top/badminton/"},{"title":"Overview of the Indian #sportstech landscape ? CoE IoT Official Blog","url":"https://www.coe-iot.com/blog/overview-of-the-indian-sportstech-landscape/"},{"title":"Top startups in Sports Tech in India (Nov, 2025) - Tracxn","url":"https://tracxn.com/d/explore/sports-tech-startups-in-india/__4UiJ6xHvc6LY7m5rZxESDNuF11gnBudqw-M12IEDuR0/companies"},{"title":"Sports Tech in India - 2025 Market & Investments Trends - Tracxn","url":"https://tracxn.com/d/explore/sports-tech-startups-in-india/__4UiJ6xHvc6LY7m5rZxESDNuF11gnBudqw-M12IEDuR0"},{"title":"The Rise Of Sports Tech: How Innovation Is Revolutionising Indian And Global Sports | IndiaSportsHub","url":"https://indiasportshub.com/the-rise-of-sports-tech-how-innovation-is-revolutionising-indian-and-global-sports/"}]

Based on the search results, here's what I found about India's sports tech startup funding in

2025, particularly related to badminton academy technology and video analysis:

Key Badminton-Focused Startups with Recent Funding

Machaxi - Founded in 2022, Machaxi operates sports training centers starting with badminton and attracted a $1.5M seed round in 2025 led by Rainmatter and Prakash Padukone himself. The system lets any player record their session via a phone camera, then uses computer vision to suggest corrections in posture, technique, etc., providing real-time personalized feedback.

VISIST.AI - A Bengaluru-based startup that uses an artificial intelligence model to analyze video footage, informing players about flaws in their posture, grip, swing velocity and overall technique. More than 70 academies in Bengaluru have been employing the technology, and VISIST.AI initiated a partnership with Padukone in late July.

Game Theory - In July 2024, the startup roped in India's chief badminton coach Pullela Gopichand and a host of other players as investors in an undisclosed funding round. The startup measures individuals' skills in sports via its computer vision-enabled smart tech courts, which record and analyse players on different levels.

Broader Market Context

The Sports Tech sector in India comprises 2.12K companies, including 197 funded companies having collectively raised $1.97B in venture capital money and private equity. The country's sports tech opportunity is projected to surpass the $3.5 Bn mark by 2027.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

19. assistant (2026-06-10T05:34:53.444Z)

The India search surfaced direct Bangalore competitors the internal docs don't mention. Let me dig deeper.

20. user (2026-06-10T05:35:01.235Z)

Web search results for query: "VISIST.AI badminton video analysis academies Bengaluru pricing"

Links: [{"title":"VISIST.AI partners with Padukone School of Badminton to digitise athlete skill assessments","url":"https://theprint.in/india/visist-ai-partners-with-padukone-school-of-badminton-to-digitise-athlete-skill-assessments/2716417/"},{"title":"Indian badminton players use artificial intelligence to up their game - Mugglehead Investment Magazine","url":"https://mugglehead.com/indian-badminton-players-use-artificial-intelligence-to-up-their-game/"},{"title":"VISIST.AI Partners with Padukone School of Badminton to Revolutionize Athlete Training | Sports-Games","url":"https://www.devdiscourse.com/article/sports-games/3533385-visistai-partners-with-padukone-school-of-badminton-to-revolutionize-athlete-training"},{"title":"Top 8 Badminton Academies in Bangalore | Best Training Centres","url":"https://www.decathlon.in/blog/Top-8-Badminton-Academies-in-Bangalore"},{"title":"Top 8 Badminton Academies in Bangalore","url":"https://blog.decathlon.in/articles/badminton-academies-in-bangalore"},{"title":"Top 10 Famous Badminton Academies in India 2025 - Sports Realize","url":"https://www.sportsrealize.com/badminton-academy-in-india/"},{"title":"Badminton Classes in Bangalore | Contact Top Academies & Trainers near you | PlayGloba","url":"https://playgloba.com/bangalore/classes/badminton"},{"title":"This Startup aims to assist badminton players with an AI coach","url":"https://indiaai.gov.in/article/this-startup-aims-to-assist-badminton-players-with-an-ai-coach"},{"title":"7 Top Badminton Academies In India: Be the Next PV Sindhu!","url":"https://school.careers360.com/articles/top-7-badminton-academies-in-india-with-cost"}]

Based on the search results, here's what I found about VISIST.AI and badminton video analysis in Bengaluru:

VISIST.AI Overview

VISIST.AI is a sports tech start-up incubated by the FSID at the Indian Institute of Science. The company uses artificial intelligence to deliver performance insights, skill assessments, and data-driven coaching for athletes and sports academies.

Deployment in Bengaluru Academies

More than 70 academies in Bengaluru have been employing the technology, many of which are overseen by the Padukone School of Badminton. VISIST.AI's Visual Intelligence platform has been deployed across PSB's academy ecosystem to enable data-driven athlete evaluations, personalised feedback, and evidence-based training insights.

Features and Capabilities

VISIST.AI's program will inform the player about flaws in their posture, grip, swing velocity and overall technique. The platform allows standardised digital assessments of key skills, AI-powered video analysis for skill benchmarking, detailed progress reports for players and parents, and rich performance data to help coaches create personalised development plans.

Pricing Information

Unfortunately, the search results do not contain any specific pricing information for VISIST.AI's services. The results mention general badminton academy pricing in Bengaluru (ranging from ?1,500 to ?5,000+ per month for group sessions), but no specific pricing for VISIST.AI's video analysis platform was found.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

21. user (2026-06-10T05:35:01.328Z)

Web search results for query: "Gemini video understanding temporal localization rally segmentation sports long video 2025 2026 capability"

Links: [{"title":"Gemini Video Understanding: Complete Multimodal AI Guide 2025","url":"https://aionx.co/gemini-advanced-reviews/gemini-video-understanding/"},{"title":"F^{3}Set: Towards Analyzing Fast, Frequent, and Fine-grained Events from   Videos","url":"https://arxiv.org/pdf/2504.08222"},{"title":"Temporal event localization in sports videos via self-supervised proposal generation and cross-modal fusion - ScienceDirect","url":"https://www.sciencedirect.com/science/article/pii/S2667305325000651"},{"title":"VideoTG-R1: Boosting Video Temporal Grounding via Curriculum Reinforcement Learning on Reflected Boundary Annotations","url":"https://arxiv.org/pdf/2510.23397"},{"title":"Video Understanding with Gemini: Notes From the Field | by Sveta Morag | Google Cloud - Community | Medium","url":"https://medium.com/google-cloud/video-understanding-with-gemini-notes-from-the-field-82dd0cd130ea"},{"title":"1 + 1 > 2: Detector-Empowered Video Large Language Model for Spatio-Temporal Grounding and Reasoning","url":"https://arxiv.org/pdf/2512.06673"},{"title":"Video-in-the-Loop: Span-Grounded Long Video QA with Interleaved Reasoning","url":"https://arxiv.org/pdf/2510.04022"},{"title":"Video understanding | Gemini Enterprise Agent Platform | Google Cloud Documentation","url":"https://docs.cloud.google.com/gemini-enterprise-agent-platform/models/capabilities/video-understanding"},{"title":"Video understanding - generateContent API","url":"https://ai.google.dev/gemini-api/docs/video-understanding"},{"title":"Advancing the frontier of video understanding with Gemini 2.5 - Google Developers Blog","url":"https://developers.googleblog.com/en/gemini-2-5-video-understanding/"}]

Based on the search results, here's what I found about Gemini's video understanding capabilities related to temporal localization and sports analysis:

Gemini's Video Understanding Capabilities (2025-2026)

Temporal Localization and Event Detection:
Gemini can identify temporal relationships and sequence of events, detect scene changes, track

objects across frames, and localize events to specific timestamps. True video understanding requires "temporal sensitivity" ? knowing not just what happened, but exactly when and why.

Long Video Support:
Gemini 1.5 Pro outperforms other models with a 1M token context window (supporting up to 1 hour of video) compared to GPT-4V's 128K tokens (5-10 minutes), with native video processing as a continuous stream rather than frame-by-frame. This is the longest native video support of any publicly available AI model.

Timestamp and Segment Analysis:
For timestamp localization in videos with audio, the model can generate timestamps, and higher FPS sampling rates are recommended for videos requiring granular temporal analysis, such as fast-action understanding or high-speed motion tracking.

Sports Video Application:
While the search results don't specifically address rally segmentation in sports videos, research has proposed methods for text snippet localization in long sports videos combining efficient retrieval with fine-grained refinement, using self-supervised proposal generation and two-stage training for segment-level and instance-level analysis.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

22. user (2026-06-10T05:35:15.024Z)

Web search results for query: "Machaxi badminton AI training centers Rainmatter Prakash Padukone funding pricing"

Links: [{"title":"Machaxi Secures $1.5 Million to Expand AI-Powered Badminton Coaching Across India","url":"https://india.entrepreneur.com/news-and-trends/machaxi-secures-15-million-to-expand-ai-powered-badminton/493070"},{"title":"Machaxi Raises $1.5M to Revolutionize Grassroots Badminton with AI and Prakash Padukone","url":"https://www.tice.news/tice-dispatch/filling-the-sports-gap-rainmatter-funds-ai-sports-startup-machaxi-9353716"},{"title":"Machaxi-Padukone School of Badminton to set up centre in Hyderabad - Telangana Today","url":"https://telanganatoday.com/machaxi-padukone-school-of-badminton-to-set-up-centre-in-hyderabad"},{"title":"Machaxi Raises $1.5M from Prakash Padukone, Rainmatter and Others to Expand AI-Based Grassroots Badminton Training - Passionate In Marketing Machaxi Raises $1.5M to Expand AI-Powered Grassroots Badminton Coaching Across India","url":"https://www.passionateinmarketing.com/machaxi-raises-1-5m-from-prakash-padukone-rainmatter-and-others-to-expand-ai-based-grassroots-badminton-training/"},{"title":"Machaxi Raises $1.5 Million to Scale AI-Powered Badminton Coaching Across India ? Outlook Business","url":"https://www.outlookbusiness.com/start-up/investors/machaxi-raises-15-million-to-scale-ai-powered-badminton-coaching-across-india"},{"title":"Machaxi Secures $1.5 Million Investment to Expand AI-Powered Badminton Coaching & Grassroots Sports Development","url":"https://startupsuccessstories.in/machaxi-secures-1-5-million-investment-to-expand-ai-powered-badminton-coaching-grassroots-sports-development/"},{"title":"Machaxi aces $1.5m fund raise from Padukone and Nithin Kamath?s Rainmatter | 1 Indian Television Dot Com","url":"https://www.indiantelevision.com/mam/digital/machaxi-aces-$1.5m-fund--raise-from-padukone-and-nithin-kamath%E2%80%99s-rainmatter-250611"},{"title":"Machaxi Raises $1.5M from Prakash Padukone, Rainmatter and Others to Expand AI-Based Grassroots Badminton Training. - Business Micro","url":"https://businessmicro.in/business/machaxi-raises-1-5m-from-prakash-padukone-rainmatter-and-others-to-expand-ai-based-grassroots-badminton-trainin/"},{"title":"Machaxi raises $1.5 Mn from Rainmatter, Prakash Padukone, others","url":"https://entrackr.com/snippets/machaxi-raises-15-mn-from-rainmatter-prakash-padukone-others-9353316"}]

Based on the search results, here's what I found about Machaxi:

Funding and Investment

Machaxi, a Bengaluru-based sports-tech startup focused on grassroots coaching, has raised $1.5 million in funding to scale its AI-powered training model and expand operations into three more cities?Hyderabad, Pune, and Chennai.

The funding round was led by Rainmatter, the investment arm of Zerodha, and included participation from Indian badminton legend Prakash Padukone as well as existing backers.

AI-Powered Badminton Training Centers

The AI coaching system being developed by Machaxi is built to empower existing coaches and enable new ones. It offers real-time performance analytics, suggested training modules based on player profiles, and consistency in the delivery of drills and feedback. This ensures that whether a player trains in a metro or a small town, they receive similar quality inputs and structured progression.

Expansion Plans

As a long-standing advocate for structured sports training, Padukone has partnered with Machaxi to co-launch a new initiative under the banner "Machaxi x Padukone School of Badminton." The initiative aims to build over 1,000 coaching centres across India over the next four years.

Note on Pricing

The search results do not contain information about pricing for Machaxi's training centers. The available sources focus on the funding announcement, expansion plans, and AI features, but do not provide specific pricing details for their training programs.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

23. user (2026-06-10T05:35:17.821Z)

Web search results for query: ""multi-court" OR "multiple courts" sports video AI camera academy gym filming problem solution"

Links: [{"title":"Sports AI and Automated Production Technology","url":"https://playsight.com/"},{"title":"Home - Spiideo","url":"https://www.spiideo.com/"},{"title":"PIX4TEAM 2 ? AI Auto-Follow Camera for Team Sports | Move'N See","url":"https://shop.movensee.com/en/products/pix4team-automatic-camera-sports.html"},{"title":"BallerCam - The AI Sports Tracking Camera for Everyone","url":"https://ballercam.com/"},{"title":"AI Camera for Sports: How It Works and Which System Is Best [2026]","url":"https://www.veo.com/article/ai-camera-for-sports-guide"},{"title":"Buy XbotGo Chameleon | The Best AI Sports Camera System","url":"https://xbotgo.com/products/xbotgo-chameleon-ai-powered-4k-sports-camera-system-phone-mount"},{"title":"Padel SmartCourt - PlaySight","url":"https://playsight.com/our-sports/padel/"},{"title":"Basketball Tournament Filming Guide 2025: 5 Pro Tips ? XbotGo","url":"https://xbotgo.com/blogs/buying-guide/basketball-tournament-recording-guide"},{"title":"Hudl Focus | AI Sports Camera with Streaming","url":"https://www.hudl.com/products/focus"},{"title":"Best Auto Tracking Camera for Basketball in 2025 Guide","url":"https://arwall.co/blogs/arwall-blogs/auto-tracking-camera-for-basketball"}]

Based on the search results, here are the key solutions for multi-court and multiple courts sports video AI camera systems for academies, gyms, and filming:

AI Camera Systems for Multiple Courts

PlaySight SmartCourt AI technology is a connected camera platform used for broadcast, instant replay and player development, delivering multiangle videos, AI data and video analysis, VAR Light, automated production, live streaming and Remote Production solutions. PlaySight has been used at championships with up to 24 cameras running at the same time.

NBA teams have implemented Spiideo's multi-angle solution to capture and analyze training sessions, providing coaches full-court visibility and players visual feedback. The Ontario Hockey League (OHL) implemented a multi-angle video capture solution across its arenas, bringing officiating, player safety, and player development up to NHL standards.

Single Court Auto-Tracking Solutions

The PIX4TEAM 2 uses a 180° camera at the base that captures the entire field, with AI deciding where to film based on player positions, the ball, and specific action recognition, while a top camera follows the action and automatically zooms in or out depending on how play unfolds.

For coaching analysis that requires seeing the full pitch or court, only wide-angle systems provide the necessary footage. Some systems include zone mapping features for basketball where you can mark the key areas on the court, providing precision that optimizes AI-tracking for full-court recording.

Tournament/Multi-Game Recording

XbotGo Chameleon delivers AI-powered auto-tracking to remove the need for a cameraman, automatically recording every match throughout a tournament day.

These solutions address the key challenges of filming multiple courts and games efficiently while maintaining professional video quality for coaching analysis and broadcasting.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

24. user (2026-06-10T05:35:36.851Z)

Web search results for query: "WSC Sports Magnifi Reely AI highlights amateur sports availability pricing"

Links: [{"title":"Automated Sports Highlights, AI Video Highlights | Magnifi","url":"https://magnifi.ai/"},{"title":"BlendVision AI vs. WSC Sports vs. Magnifi: Which AI Video Solution Is Right for Your Media Workflow?","url":"https://blendvision.com/en/blog/top-ai-video-solutions-compared-which-one-fits-your-media-strategy/"},{"title":"WSC Sports","url":"https://wsc-sports.com/"},{"title":"WSC Sports: How AI Has Led the Highlights Revolution","url":"https://www.sportcal.com/features/wsc-sports-how-ai-has-led-the-highlights-revolution/"},{"title":"WSC Sports - YouTube","url":"https://www.youtube.com/@wsc-sports"},{"title":"Reely | Magnifi","url":"https://magnifi.ai/products/reely"},{"title":"WSC Sports vs Magnifi.AI: Video Analytics Comparison","url":"https://6sense.com/tech/video-analytics/wscsports-vs-magnifiai"},{"title":"AI Sports Revolution: 12 Innovations Changing Everything - WSC Sports","url":"https://wsc-sports.com/blog/industry-insights/ai-sports-revolution-12-innovations-changing-everything/"},{"title":"How WSC Sports Uses Wowza to Automate AI-Driven Sports Highlights","url":"https://www.wowza.com/blog/case-study-wsc"},{"title":"AWS Marketplace: WSC Sports AI-powered platform (Private Offers Only)","url":"https://aws.amazon.com/marketplace/pp/prodview-k7i7gdfrvcpjw"}]

Based on the search results, here's what I found about these AI sports highlight platforms:

WSC Sports

WSC Sports is an AI-powered platform that helps leagues, broadcasters, and media partners turn every moment into personalized highlights?automated, real-time, and built for fan engagement. 550+ organizations including the NBA, NHL, LALIGA, and DAZN trust WSC to scale content and maximize impact.

For organizations focused purely on sports content, WSC delivers reliable results. However, WSC and Magnifi are less forthcoming about cost structure, which may complicate budgeting for tech buyers.

Magnifi and Reely

Magnifi's Reely product elevates esports and gaming content with AI-generated highlight clips, and while you focus on gameplay, Reely generates high-quality, personalized clips from your broadcast in real-time. The platform supports multiple top-tier platforms like Twitch, YouTube, and Trovo, giving users maximum flexibility.

Availability for Amateur Sports

Reely's automated sharing solution is designed for professional esports athletes or casual gamers to showcase their highlights and engage with their audience on preferred channels. The results suggest Reely is particularly focused on esports and gaming content rather than traditional amateur sports.

Pricing Transparency

Unfortunately, the search results indicate that neither WSC Sports nor Magnifi provides transparent public pricing information on their websites. Both platforms appear to use custom enterprise pricing models based on specific organizational needs and contract terms.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

25. user (2026-06-10T05:35:37.322Z)

Gemini 2.5 Video Understanding: Temporal Capabilities & Limits

Temporal Capabilities

The blog demonstrates temporal localization through specific examples:

- **Moment retrieval**: Gemini 2.5 Pro "accurately identifies 16 distinct segments related to product presentations, using both audio and visual cues" from a 10-minute video.

- **Temporal reasoning/counting**: The model successfully counted "17 distinct occurrences where the main character uses their phone" throughout a video, showing ability to track events across time.

Benchmark Performance

Specific metrics mentioned include:
- QVHighlights moment retrieval benchmark (measured by R1@0.5)
- YouCook2 dense captioning (measured by CIDEr)
- VideoMME accuracy: "84.7% vs 85.2% accuracy"

The blog notes Gemini 2.5 Pro "rivals specialized fine-tuned models on several challenging benchmarks."

Technical Limits

Frame/length constraints:
- "Videos were processed at 1fps and linearly subsampled to a maximum of 256 frames"
- Exception: "1H-VideoQA (7200 frames)"
- With the 'low' resolution parameter, the API can process "~6 hours of video with 2 million token context"

Not Addressed

The blog contains no specific information about sports use cases or general long-video segmentation performance beyond the examples shown.

26. user (2026-06-10T05:35:52.730Z)

Web search results for query: "Veo camera pricing 2026 plans cost per year club"

Links: [{"title":"Google Veo Pricing Calculator & Cost Guide (Jun 2026)","url":"https://costgoat.com/pricing/google-veo"},{"title":"Plans &

Pricing","url":"https://www.veo.com/pricing"},{"title":"VEO Camera: our complete guide | SoccerEDU","url":"https://www.socceredu.com/en-US/blog/veo-camera"},{"title":"Spring Season Sale - 2026 | Veo","url":"https://www.veo.com/en-us/spring-season-2026"},{"title":"Overview of Veo subscription plans: features and entitlements included ? Veo Help Center","url":"https://support.veo.com/hc/en-us/articles/4450464075025-Overview-of-Veo-subscription-plans-features-and-entitlements-included"},{"title":"Veo 3 Pricing Plans Explained: Subscription Tiers, Hidden Costs & Budget Tips | Breaking AC","url":"https://breakingac.com/news/2026/feb/18/veo-3-pricing-plans-explained-subscription-tiers-hidden-costs/"},{"title":"What Is a Veo Camera for Soccer? Full Comparison & Buying Guide","url":"https://electronics.alibaba.com/question/veo-soccer-camera-guide-cost,-setup-real-world-use"},{"title":"Veo Camera Price & Plan Costs - Soccer Stripes","url":"https://soccerstripes.com/veo-camera-price-plan-costs/"}]

Based on the search results, here's what I found about Veo camera pricing for 2026:

Camera Hardware Costs

The Veo Cam 3 starts at about $1,533 (one-time payment), while the Veo Cam 3 5G starts at about $1,886 (one-time payment).

Subscription Plans for Clubs

Veo offers multiple subscription tiers:

- The "Team" plan costs $49/month and adds unlimited player profiles, advanced tagging, and live-streaming
- The "Club" plan costs $149/month and enables multi-camera synchronization and API access for third-party analytics platforms

Optional Add-ons

Veo Live starts at $12 per month, while Veo Analytics provides advanced soccer data powered by AI, starting at $26 per month.

Flexible Payment Options

You can pay monthly, every 6 months, annually, or every 2 years ? the longer the plan, the lower the cost per month.

The exact total cost will depend on which camera model you choose, which subscription tier suits your club's needs, and whether you opt for additional features like analytics or livestreaming capabilities.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

27. user (2026-06-10T05:35:54.904Z)

Web search results for query: "how many badminton academies in India Khelo India badminton participation growth"

Links: [{"title":"#1 Badminton in India","url":"https://mjsportsacademy.in/badminton-in-india-growth-stars-of-future/"},{"title":"Kheloindia","url":"https://kheloindia.gov.in/uploads/Khelo_India_Accrediated_Academies.pdf"},{"title":"Increased participation to global podium finishes: India's sporting transformation between 2014-2025","url":"https://www.prokerala.com/news/articles/a1642894.html"},{"title":"Increased participation to global podium finishes: India's sporting transformation between 2014-2025 - India Tribune - Chicago","url":"http://www.indiatribune.com/increased-participation-to-global-podium-finishes-indias-sporting-transformation-between-2014-2025"},{"title":"Khelo India Programme: Grassroots Sports & Infrastructure in India","url":"https://gallantsports.in/khelo-india-programme"},{"title":"Khelo India: Building India?s Sporting Future From The Grassroots | IndiaSportsHub","url":"https://indiasportshub.com/khelo-india-building-indias-sporting-future-from-the-grassroots/"},{"title":"Badminton","url":"https://kheloindia.gov.in/uploads/Circular%20_%20Badminton.pdf"},{"title":"Khelo India Scheme","url":"https://cleartax.in/s/khelo-india-scheme"},{"title":"The Evolution of Badminton in India: From Grassroots to Global Stage ? 333k+ Inspiring Lifestyle Ideas","url":"https://artfasad.com/the-evolution-of-badminton-in-india-from-grassroots-to-global-stage/"},{"title":"Khelo India Programme: Objectives and Verticals","url":"https://getswipe.in/blog/article/khelo-india-programme-objectives-verticals"}]

Based on the search results, here's what I found about badminton academies in India and Khelo India's badminton participation growth:

Badminton Academies in India

While the search results don't provide a specific total number of all badminton academies in India, they do indicate significant academy development:

The Khelo India program has accredited 306 academies across various sports disciplines. Private academies have established themselves as nurseries of talent, with the Pullela Gopichand Badminton Academy having made champions with Saina Nehwal and P.V. Sindhu. Additionally, several academies, like the Prakash Padukone Badminton Academy and Pullela Gopichand Academy, have provided top-notch training to players.

Khelo India and Badminton Participation Growth

The expansion of badminton in India is due in a large part to organized efforts by the government and the establishment of top quality training academies, with the Target Olympic Podium Scheme (TOPS), Khelo India, and public and private partnerships developing an excellent support system for elite level athletes and young athletes.

The Khelo India program established 1,045 Khelo India Centres (KICs) for grassroots training and support, and notified 34 Khelo India State Centres of Excellence (KISCEs). Badminton is included as one of the competitive disciplines in the Khelo India Games.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

28. user (2026-06-10T05:36:07.265Z)

Web search results for query: "Game Theory app India racket sports courts video recording booking Gopichand"

Links: [{"title":"Sports Membership & Coaching in Bangalore - Game Theory","url":"https://www.gametheory.in/all-old-pages/old-home"},{"title":"Game Theory India ? Apps on Google Play","url":"https://play.google.com/store/apps/details?id=in.oodadu.gametheory&hl=en_SG"},{"title":"Game Theory India App - App Store","url":"https://apps.apple.com/in/app/game-theory-india/id1489081484"},{"title":"Game Theory - Crunchbase Company Profile & Funding","url":"https://www.crunchbase.com/organization/game-theory-3df5"},{"title":"Game Theory (@gametheoryindia)","url":"https://www.instagram.com/gametheoryindia/?hl=en"},{"title":"Game Theory - Products, Competitors, Financials, Employees, Headquarters Locations","url":"https://www.cbinsights.com/company/game-theory"},{"title":"Game Theory aims to facilitate more than 3,20,000 sports enthusiasts through their mobile app | NotInTown","url":"https://www.notintown.net/india-details/1562/game-theory-aims-to-facilitate-more-than-3-20-000-sports-enthusiasts-through-their-mobile-app.html"},{"title":"GameTheory - 2026 Company Profile, Team, Funding, Competitors & Financials - Tracxn","url":"https://tracxn.com/d/companies/gametheory/__pdOB46xo2J1QH-rpu1vpiw-gl-kIJXu3VNWhc9otULM"},{"title":"Game Theory - YouTube","url":"https://www.youtube.com/@gametheoryindia"},{"title":"Using computer vision, Game Theory helps sports players ...","url":"https://yourstory.com/2024/02/computer-vision-game-theory-sports-tech-startup-nithin-kamath-rainmatter"}]

Based on the search results, here's what I found about Game Theory:

Overview

Game Theory is a company that operates in the sports and fitness industry and is based in Bengaluru, India, founded in 2018.

Core Features

Game Theory is a mobile application for booking sports courts, accessing coaching programs, joining sports communities, and tracking sports activities. Their offerings include video highlights of sports activities, smart courts for real-time scoring and skill tracking, and a platform for skill-based matchmaking among players.

Pullela Gopichand Connection

Game Theory's programs are created by champions like Pullela Gopichand, Rohan Bopanna, Sharath Kamal, and Saurav Ghosal ? elite athletes who know what it takes to rise to the top. Additionally, Pullela Gopichand, Sharath Kamal, and Saurav Ghosal are investors of Game Theory.

Video Recording & Booking

The app provides Instagram-ready video highlights that users can access after their game through the Game Theory app. Users can book courts, train with the best, and join a thriving sports community through the Game Theory app.

Sports Covered

Game Theory offers Swimming, Badminton, Basketball, Squash, Pickleball, Padel, Football, and Table Tennis.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

29. user (2026-06-10T05:36:09.168Z)

Web search results for query: "padel video analytics startup 2025 AI camera court Wingfield padel highlights"

Links: [{"title":"Wingfield AI Sports Cameras | Video & Tracking Solutions for Padel, Tennis & Pickleball","url":"https://www.wingfield.io/en"},{"title":"Clutch - The AI Camera for Padel | Automated Highlights & Analytics","url":"https://www.clutchapp.io/"},{"title":"Wingfield AI Sport Cameras: Video & Tracking for Racket Sports","url":"https://www.navgood.com/en/tool-details/wingfield-7e633"},{"title":"Wingfield Video Analytics System: Sports Video Analysis Tool & Automated Highlight Creator | Find Me AI","url":"https://findmeai.com/tools/5M1q/wingfield-video-analytics-system"},{"title":"Can AI Make You Better at Padel? Smart Training, Video Analysis and th ? Padelspeed","url":"https://padelspeed.com/blogs/news/can-ai-make-you-better-at-padel-smart-training-video-analysis-and-the-future-of-coaching"},{"title":"Home | Padelytics","url":"https://www.padelytics.ai/"},{"title":"Padel SmartCourt - PlaySight","url":"https://playsight.com/our-sports/padel/"},{"title":"Wingfield Video Suite | Video Tools for Padel, Tennis & Pickleball","url":"https://www.wingfield.io/en/products/video-suite"},{"title":"Wingfield video tracking for tennis, pickleball, and padel courts","url":"https://www.messe.tv/en/2025/ispo/wingfield-video-tracking-for-tennis-pickleball-and-padel-courts"},{"title":"Wingfield Sports - Apps on Google Play","url":"https://play.google.com/store/apps/details?id=com.wingfield"}]

Based on the search results, here's what I found about padel video analytics startups with AI camera and court technology in 2025:

Key Players in Padel AI Video Analytics

Wingfield
Wingfield is an AI camera solution for tennis, padel & pickleball offering video analysis, automatic tracking, highlights & streaming in one system. Wingfield's AI evaluates every rally and automatically generates highlight reels of your most spectacular moments. Renowned academies such as the Rafael Nadal Academy are already using the technology.

Clutch
Clutch turns every game into highlights, rankings, and performance insights with a new open

rating system built for padel. The AI tracks each individual player, their movements, and every shot.

Padelytics
Padelytics transforms your video platform into a padel intelligence hub. By analyzing every frame, they track players and the ball, detecting and classifying player actions, and generating a comprehensive event stream of everything that happens on the court.

PlaySight
PlaySight offers an all-in-one multi-angle video platform for broadcast, replay (VAR), coaching, and automated highlights with AI-powered video editing tools.

Core Features

These platforms share common capabilities: AI tools and smart cameras are designed to notice the quieter patterns that players often miss, tracking rallies, recording matches, generating highlights, analysing shot types and helping players review performance without watching a full two-hour match manually.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

30. user (2026-06-10T05:36:28.282Z)

Web search results for query: "SwingVision badminton padel support roadmap adding new sports"

Links: [{"title":"SwingVision FAQ | Tennis & Pickleball App Support","url":"https://swing.vision/faq"},{"title":"SwingVision on Wefunder 2025 - Kingscrowd","url":"https://kingscrowd.com/swingvision-on-wefunder-2025/"},{"title":"SwingVision Raises $6M to Reach Pickleball and Padel - Profluence","url":"https://profluence.com/swingvision-raises-6m-to-reach-pickleball-and-padel/"},{"title":"AI tennis app SwingVision raises US$6m to expand into ...","url":"https://www.sportspro.com/news/swingvision-ai-tennis-app-pickleball-padel/"},{"title":"SwingVision | AI Tennis & Pickleball App (@SwingVisionApp) / X","url":"https://x.com/swingvisionapp"},{"title":"SwingVision (@swingvisionapp) ? Instagram photos and videos","url":"https://www.instagram.com/swingvisionapp/"},{"title":"SwingVision: AI Stats for Tennis & Pickleball","url":"https://swing.vision/"}]

Based on the search results, here's what I found about SwingVision's plans for badminton, padel, and other sports:

SwingVision's technology for tracking fast-moving balls and players could be applied to other racket sports like padel, squash, or badminton, and expansion to padel is on the roadmap. Additionally, expansion into other racket sports, including padel, is also on the company's roadmap.

Currently, SwingVision's core product today primarily serves the tennis and pickleball segment of this market. The company has ambitious growth plans, as the company's ultimate goal is to create a multi-sport platform that redefines how athletes of all levels record, share, and enhance their game experiences.

However, it's worth noting that pushing into new sports such as pickleball or padel introduces execution risk, as developing and validating sport-specific AI while marketing to new communities could stretch resources and compromise product quality.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

31. user (2026-06-10T05:36:29.770Z)

Web search results for query: "Stupa Sports Analytics India table tennis badminton funding pricing"

Links: [{"title":"Stupa Sports Analytics","url":"https://stupasports.ai/"},{"title":"Stupa Sports Analytics Nets INR 28 Cr To Offer AI Solutions To Sports Federations","url":"https://inc42.com/buzz/stupa-sports-analytics-nets-inr-28-cr-to-offer-ai-solutions-to-sports-federations/"},{"title":"Badminton, Padel, Table Tennis to use AI driven IRS after BWF Approval: WTT, India Open set for New Review Procedure - myKhel","url":"https://www.mykhel.com/more-sports/badminton-padel-table-tennis-to-use-ai-driven-irs-after-bwf-approval-wtt-india-open-set-for-new-420431.html"},{"title":"How data analytics is smashing it in Indian table tennis | SME Futures | SME News India","url":"https://smefutures.com/how-data-analytics-is-smashing-it-in-indian-table-tennis/"},{"title":"Sportstech startup Stupa Sports Analytics raises Rs 28Cr in a pre-Series A round","url":"https://indianstartupnews.com/funding/sportstech-startup-stupa-sports-analytics-raises-rs-28cr-in-a-pre-series-a-round-4581092"},{"title":"Stupa Sports Analytics Secures Rs 280 Million in Pre-Series A Funding Round ? Outlook Business","url":"https://www.outlookbusiness.com/corporate/stupa-sports-analytics-secures-rs-280-million-in-pre-series-a-funding-round"}]

Based on the search results, here's information about Stupa Sports Analytics:

Overview
Stupa Sports Analytics was founded in 2020 by Megha Gambhir and Deepak Malik and offers AI-driven solutions like player performance analytics to sports federations to help them digitise their operations and management.

Sports Coverage
Table tennis dominates the company's services. However, the company started with table tennis and has now expanded to badminton, with padel, pickleball, and squash to follow soon.

Funding
Stupa Sports Analytics secured INR 28 Cr (approximately $3.3 Mn) as a part of its Pre-Series A funding round co-led by Centre Court Capital and PeerCapital. Additionally, between 2020 and 22, the company raised around INR 7.5 Cr (approximately $898,227) in multiple angel rounds from micro VCs and angel investors, including Akshat Rathee, Sanjeev Shah and Bharat founders fund, among others.

Products and Services
Stupa's platform provides real-time insights into things like shot placement, serve patterns, and rally length, which coaches and players use to tweak strategies mid-game. The company provides fan engaging content for TV broadcasters, majorly catering to Tier 2 sports like badminton, padel, pickleball, volleyball and rugby, among others.

Partnerships
Stupa is currently catering to more than 15 sports federations.

Note: The search results do not contain specific pricing information for Stupa's services.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

32. user (2026-06-10T05:36:45.256Z)

Based on the provided content, I can offer limited information:

What is F^3Set:
F^3Set is described as "a benchmark that consists of video datasets for precise F^3 event detection," focusing on Fast, Frequent, and Fine-grained events. The datasets feature "over 1,000 event types with precise timestamps and supporting multi-level granularity."

Sports Coverage:
The abstract states that "F^3Set contains several sports datasets," but the specific sports included are not detailed in this excerpt. The content mentions "sports datasets" without naming them explicitly.

Performance Results:
Unfortunately, the provided webpage content does not include concrete performance metrics or results. The abstract notes that researchers "evaluated popular temporal action understanding methods on F³Set, revealing substantial challenges for existing techniques" and that a new method called F³ED achieved "superior performance," but no specific numbers, comparisons with GPT-4o, Gemini, Qwen-VL, or other VLMs are provided in this excerpt.

To obtain the detailed performance data you're seeking, you would need to access the full paper PDF or the project website mentioned in the abstract.

33. user (2026-06-10T05:36:46.760Z)

Web search results for query: "Trace soccer camera subscription price 2026 traceup"

Links: [{"title":"Trace Pricing: Free Camera + Subscription | Start Today","url":"https://traceup.com/pricing"},{"title":"Trace: The Ultimate Soccer Camera","url":"https://traceup.com/"},{"title":"How Trace Subscriptions Work: A Quick Guide","url":"https://traceup.com/academy/how-trace-subscriptions-basic-vs-pro-plans"},{"title":"Trace Soccer Camera System ? Legends FC","url":"https://www.legendsfc.net/trace-soccer-camera-system"},{"title":"TraceCam | AI Sports Camera with Automatic Highlights","url":"https://traceup.com/soccer-camera"},{"title":"Amazon.com: Trace Camera Soccer","url":"https://www.amazon.com/trace-camera-soccer/s?k=trace+camera+soccer"},{"title":"VEO vs. Trace: Soccer Camera Review","url":"https://www.recordingsports.com/post/veo-trace-soccer-camera-review"},{"title":"Washington Youth Soccer and Trace Partnership","url":"https://traceup.com/partner/wys"},{"title":"Minnesota Youth Soccer and Trace Partnership","url":"https://traceup.com/partner/mysa"},{"title":"Does anyone know Club Level pricing for the Trace ...","url":"https://www.facebook.com/groups/guestplayers/posts/4153566224966249/"}]

Based on the search results, here's the Trace soccer camera subscription pricing for 2026:

Subscription Plans

Trace offers a Basic plan at $180/year or $25/month with 2 seats, and a Pro plan at $300/year or $35/month with 4 seats.

Plan Features

Basic: Includes watching highlights, full games, and the last 5 matches with automatic spotlight on your player, plus personalized highlights in PlayerCam with 4x zoom or tactical views.

Pro: Includes everything in Basic plus access to ALL games, download highlights, create unlimited playlists, and enjoy player heatmaps and individual and team analytics.

Camera Lease

With a Trace subscription, you get a free AI camera and tripod, so no additional hardware purchase is required?just the subscription cost.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

34. user (2026-06-10T05:37:01.395Z)

BadPro+ Overview

Based on the provided content, I can only confirm that BadPro+ is described as **"AI-Powered Visual Intelligence for Badminton"** from the page title.

Unfortunately, the web page content provided is extremely limited?containing only the title. The excerpt does not include:

- **Features**: No information about shot detection, highlights, heatmaps, video analysis, or rally segmentation
- **Pricing model**: No details on pay-per-use, subscription options, or actual prices
- **Platforms**: No platform specifications mentioned
- **Target audience**: No clarity on whether it serves players, coaches, federations, or all three
- **Highlight capabilities**: No details about automatic rally segmentation or highlight reel generation

To get this information, you would need to access the full BadPro+ website, including their features page, pricing section, and product documentation.

35. user (2026-06-10T05:37:07.811Z)

Web search results for query: "video LLM temporal grounding benchmark 2025 limitations precise boundaries moment retrieval R@0.7"

Links: [{"title":"GitHub - iLearn-Lab/TPAMI26-Awesome-MLLMs-for-Video-Temporal-Grounding: Latest Papers, Codes and Datasets on VTG-LLMs. · GitHub","url":"https://github.com/ki-lw/Awesome-MLLMs-for-Video-Temporal-Grounding"},{"title":"TimeExpert: An Expert-Guided Video LLM for Video Temporal Grounding","url":"https://arxiv.org/pdf/2508.01699"},{"title":"Enrich and Detect: Video Temporal Grounding with Multimodal LLMs","url":"https://arxiv.org/pdf/2510.17023"},{"title":"LeAdQA: LLM-Driven Context-Aware Temporal Grounding for Video Question Answering","url":"https://arxiv.org/html/2507.14784v2"},{"title":"TimeScope: Towards Task-Oriented Temporal Grounding In Long Videos","url":"https://arxiv.org/pdf/2509.26360"},{"title":"TimeRefine: Temporal Grounding with Time Refining Video LLM","url":"https://arxiv.org/pdf/2412.09601"},{"title":"Time-R1: Post-Training Large Vision Language Model for Temporal Video Grounding","url":"https://arxiv.org/pdf/2503.13377"},{"title":"DaMO: A Data-Efficient Multimodal Orchestrator for Temporal Reasoning with Video LLMs","url":"https://arxiv.org/pdf/2506.11558"},{"title":"VideoExpert: Augmented LLM for Temporal-Sensitive Video Understanding","url":"https://arxiv.org/html/2504.07519v1"}]

Based on the search results, here's what I found about video LLM temporal grounding benchmarks and their limitations regarding precise boundaries and moment retrieval:

Key Limitations in Video LLM Temporal Grounding

Most top-performing models focus primarily on overall content comprehension, lacking the ability to identify boundaries and dynamic relationships among events. More specifically, this limitation hampers their performance in temporal-sensitive tasks, which require precise moment boundaries to pinpoint when specific events occur, such as temporal grounding, highlight detection, dense video captioning, and grounding question answering.

A fundamental challenge is that while humans can effortlessly recognize actions visually, determining their precise temporal boundaries?down to the specific second?remains significantly challenging.

Despite efficiency gains, current methods still struggle with long-range dependencies and precise alignment.

Recent Approaches Addressing Precision Issues

Recent approaches like VTimeLLM introduce boundary-aware architectures designed explicitly for fine-grained temporal localization, while Momentor and ChatVTG propose specialized training objectives targeting precise moment retrieval and temporal question answering.

The search results show that efficient visual feature handling is essential for capturing more fine-grained temporal cues without overwhelming the model.

I found several 2025 papers addressing these challenges (TimeExpert, TimeScope, Time-R1, and

TimeSuite), though the specific R@0.7 metric and detailed benchmark limitations you asked about weren't fully detailed in these results. For more specific information about R@0.7 metric performance and detailed benchmark comparisons, you may want to check the full papers on arXiv directly.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

36. user (2026-06-10T05:37:19.122Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

37. user (2026-06-10T05:37:25.403Z)

Gemini Flash Models Pricing Summary

Gemini 2.5 Flash
Standard tier (paid):
- Input: "$0.30 (text / image / video)" and "$1.00 (audio)"
- Output: "$2.50"

Batch tier (paid):
- Input: "$0.15 (text / image / video)" and "$0.50 (audio)"
- Output: "$1.25"

Gemini 3 Flash Preview
Standard tier (paid):
- Input: "$0.50 (text / image / video)" and "$1.00 (audio)"
- Output: "$3.00"

Batch tier (paid):
- Input: "$0.25 (text / image / video)" and "$0.50 (audio)"
- Output: "$1.50"

Gemini 2.5 Flash-Lite
Standard tier (paid):
- Input: "$0.10 (text / image / video)" and "$0.30 (audio)"
- Output: "$0.40"

Batch tier (paid):
- Input: "$0.05 (text / image / video)" and "$0.15 (audio)"
- Output: "$0.20"

Video Token Details
The documentation does not specify a separate tokens-per-second rate for video. Video input is billed at the same per-token rate as image input across these models.

38. user (2026-06-10T05:37:38.738Z)

Web search results for query: "SwingVision Wefunder revenue ARR subscribers 2025 raise"

Links: [{"title":"SwingVision on Wefunder 2025 - Kingscrowd","url":"https://kingscrowd.com/swingvision-on-wefunder-2025/"},{"title":"SwingVision: Automated Stats, Highlights & Officiating Using Just Your Phone | Wefunder, Home of the Community Round","url":"https://wefunder.com/swingvisioncf2025"},{"title":"SwingVision on Wefunder 2024 - Kingscrowd","url":"https://kingscrowd.com/swingvision-on-wefunder-2024/"},{"title":"SwingVision on Wefunder ? Reg CF Investment Crowdfunding Raise Feed","url":"https://blog.seedstage.ai/2025/08/19/swingvision/"},{"title":"Annual Recurring

Revenue (ARR) - Wefunder","url":"https://help.wefunder.com/basic-jargon/annual-recurring-revenue-(arr)"},{"title":"SwingVision Stock Price, Funding, Valuation, Revenue & Financial Statements","url":"https://www.cbinsights.com/company/swingvision/financials"},{"title":"Swing Vision - 2025 Company Profile, Team, Funding & ...","url":"https://tracxn.com/d/companies/swing-vision/__mtjZ2VObNzXqkMo6dqLL9Cod9rbXNENVpAULBpYUGHk"},{"title":"Swing Vision - 2025 Funding Rounds & List of Investors - Tracxn","url":"https://tracxn.com/d/companies/swing-vision/__mtjZ2VObNzXqkMo6dqLL9Cod9rbXNENVpAULBpYUGHk/funding-and-investors"},{"title":"Wefunder - Crunchbase Company Profile & Funding","url":"https://www.crunchbase.com/organization/wefunder"}]

Based on the search results, here's the information about SwingVision's Wefunder raise and key metrics for 2025:

Revenue & Subscribers

SwingVision has more than 20,000 paying subscribers and more than $4 million in annual recurring revenue. In fiscal year 2024, SwingVision recorded $2,746,593 in revenue, a year-over-year increase of about 12% from $2,455,089 in 2023.

2025 Wefunder Raise

The crowdfunding campaign has a minimum target of $50,000 and a maximum target of $618,000. The campaign proceeds will be used for product development and marketing. The goal of this raise is to maximize the number of active users who can invest, rather than the amount of capital raised.

Prior Funding

Before the current Wefunder campaign, SwingVision raised over $10 million through venture rounds and prior crowdfunding. Earlier funding came from sources such as Techstars and strategic backers like Sony Innovation Fund and Tennis Australia.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

39. user (2026-06-10T05:40:40.586Z)

Structured output provided successfully

40. assistant (2026-06-10T05:40:49.576Z)

Structured output delivered. The key finding: the internal pricing and cost benchmarks all verified against live sources, but the "India badminton whitespace" claim is falsified ? Bangalore hosts at least four funded, celebrity-distributed badminton AI players (VISIST.AI, Machaxi, Game Theory, Stupa) absent from `docs/PMF_MARKET_RESEARCH.md`'s competitive table, while the multi-court software gap from `docs/NEXT_STEPS.md` emerged as the strongest genuinely unaddressed wedge.